

Simplifying an Expression, group *Alpha* α -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-6(-7x - 1) - 3(-2 - 2x)$??? $48x + 12$

(b) $3(-1 + 4x)(-5x + 1) - 1(5 - 4x)$??? $-60x^2 + 31x - 8$

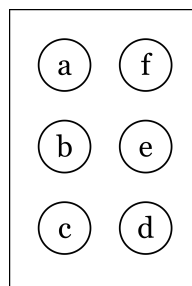
(c) $(2x - 4)^3 - (3x + 3)^2$??? $-8x^3 - 57x^2 - 73$

(d) $\frac{-3x + 7}{-2} - 3 \cdot \frac{-5 + 3x}{5}$??? $\frac{-3x + 5}{10}$

(e) $\frac{\frac{1}{-1} - \frac{1}{x}}{\frac{1}{-2} + \frac{2}{2}}$??? $\frac{-3x - 3}{2x}$

(f) $\frac{(-4 + x)^2 - 5}{(-5x + 2) \cdot \frac{-6}{x}}$??? $\frac{x^3 - 8x^2 + 11x}{30x - 12}$

1.



Braille letter

Simplifying an Expression, group *Alpha* α -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
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 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $4(4x - 1) + 3(-3 + 5x)$??? $31x - 13$

(b) $-4(-3 - 3x)(-4x - 8) - 3(-3 - 8x)$??? $-48x^2 + 120x + 87$

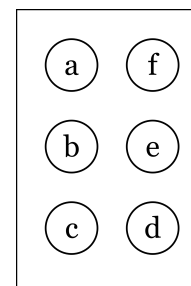
(c) $(-4x - 2)^3 - (4x - 3)^2$??? $64x^3 - 112x^2 - 17$

(d) $\frac{-5x - 2}{4} - 3 \cdot \frac{-7 - x}{-5}$??? $\frac{37x + 94}{20}$

(e) $\frac{\frac{-4}{-4} - \frac{8}{x}}{\frac{1}{-3} + \frac{2}{-1}}$??? $\frac{-15x + 97}{28x}$

(f) $\frac{(-5 + 2x)^2 - 1}{(-3x + 1) \cdot \frac{4}{x}}$??? $\frac{4x^3 - 20x^2 - 24x}{12x + 4}$

2.



Braille letter

Simplifying an Expression, group *Alpha* α -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-5(-3x + 2) + 8(2 - 3x)$??? $-9x + 6$

(b) $-2(3 + 5x)(2x - 5) + 2(-2 - x)$??? $-20x^2 - 36x - 26$

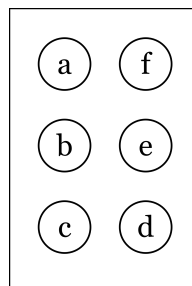
(c) $(x - 2)^3 - (3x + 2)^2$??? $-x^3 + 15x^2 - 12$

(d) $\frac{x + 8}{-4} + 6 \cdot \frac{-3 + 3x}{-9}$??? $\frac{81x}{-36}$

(e) $\frac{\frac{5}{5} - \frac{-1}{x}}{\frac{1}{-1} + \frac{4}{-2}}$??? $\frac{11x + 11}{-30x}$

(f) $\frac{(-6 - 2x)^2 + 1}{(-2x + 2) \cdot \frac{4}{x}}$??? $\frac{4x^3 + 24x^2 + 37x}{-8x + 8}$

3.



Braille letter

Simplifying an Expression, group *Alpha* α -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $2(4x - 3) + 7(-1 - 4x)$??? $-20x - 13$

(b) $-3(2 + 2x)(-7x + 4) - 1(4 - 2x)$??? $42x^2 - 20x - 28$

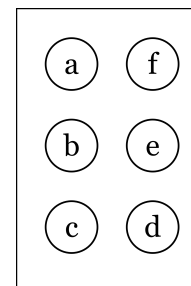
(c) $(x - 2)^3 - (-4x + 5)^2$??? $-x^3 - 22x^2 - 33$

(d) $\frac{-2x + 1}{4} + 3 \cdot \frac{-2 - 8x}{-8}$??? $\frac{-80x - 32}{32}$

(e) $\frac{\frac{-6}{3} - \frac{5}{x}}{\frac{1}{2} + \frac{1}{-3}}$??? $\frac{36x + 90}{-3x}$

(f) $\frac{(-8 + 4x)^2 - 9}{(4x - 7) \cdot \frac{3}{x}}$??? $\frac{16x^3 - 64x^2 - 55x}{-12x - 21}$

4.



Braille letter

Simplifying an Expression, group *Beta* β -i

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- (a) Expand the brackets and simplify the expression.

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(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $6(8x - 4) + 2(-8 - 3x)$??? $42x - 40$

(b) $-2(4 - 4x)(-6x + 2) - 8(-1 + 7x)$??? $-48x^2 - 8x$

(c) $(x + 1)^3 - (-6x + 4)^2$. ??? . $x^3 - 33x^2 + 51x - 15$

(d) $\frac{-5x + 3}{3} - 3 \cdot \frac{3 + x}{6}$??? $\frac{39x - 9}{-18}$

(e) $\frac{\frac{3}{4} - \frac{-1}{x}}{\frac{1}{2} + \frac{3}{3}}$??? $\frac{18x + 24}{36x}$

(f) $\frac{(6 + 6x)^2 - 9}{(-7x + 8) \cdot \frac{9}{x}}$??? $\frac{36x^3 + 72x^2 + 27x}{-63x + 72}$

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Braille letter

Simplifying an Expression, group *Beta* β -ii

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- (a) Expand the brackets and simplify the expression.

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(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $-8(4x + 5) + 5(2 + 3x)$??? $-17x - 30$

(b) $6(1 - 4x)(-4x + 3) + 4(1 - 3x)$??? $96x^2 + 108x + 22$

(c) $(-3x + 4)^3 - (3x - 3)^2$??? $27x^3 - 99x^2 - 126x + 55$

(d) $\frac{-3x + 3}{-3} + 2 \cdot \frac{-4 - 4x}{-2}$??? $\frac{18}{-6}$

(e) $\frac{\frac{-1}{3} - \frac{-2}{x}}{\frac{1}{4} + \frac{-4}{-2}}$??? $\frac{7x - 45}{-54x}$

(f) $\frac{(5 - 5x)^2 + 1}{(3x - 5) \cdot \frac{-8}{x}}$??? $\frac{25x^3 - 50x^2 - 26x}{24x + 40}$

2.

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Braille letter

Simplifying an Expression, group *Beta* β -iii

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(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $-3(2x - 8) - 7(1 + 2x)$??? $-20x + 17$

(b) $2(-3 + x)(5x + 4) + 1(3 - 8x)$... ??? ... $10x^2 + 30x$

(c) $(-2x - 3)^3 - (-2x - 2)^2$??? $-8x^3 - 40x^2 - 62x - 31$

(d) $\frac{-2x + 1}{2} + 4 \cdot \frac{-1 + 7x}{4}$??? $\frac{48x - 4}{-8}$

(e) $\frac{\frac{-1}{2} - \frac{3}{x}}{\frac{1}{1} + \frac{3}{3}}$??? $\frac{-4x - 21}{12x}$

(f) $\frac{(-1 + 2x)^2 + 4}{(-x + 2) \cdot \frac{-8}{x}}$??? $\frac{4x^3 - 4x^2 + 5x}{8x - 16}$

3.

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Braille letter

Simplifying an Expression, group *Beta* β -iv

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(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $7(3x + 2) - 1(3 + 3x)$??? $18x + 11$

(b) $2(-1 - 5x)(-2x + 2) - 3(-6 + 6x)$??? $20x^2 + 34x - 14$

(c) $(x + 4)^3 - (-6x - 1)^2$... ??? ... $-x^3 + 24x^2 + 36x + 63$

(d) $\frac{3x - 2}{-2} + 7 \cdot \frac{-6 + 7x}{5}$??? $\frac{-83x + 74}{10}$

(e) $\frac{\frac{-1}{-2} - \frac{7}{x}}{\frac{1}{-5} + \frac{-3}{1}}$??? $\frac{5x - 70}{-32x}$

(f) $\frac{(-7 + 5x)^2 + 5}{(-x + 2) \cdot \frac{4}{x}}$??? $\frac{25x^3 - 70x^2 - 54x}{4x + 8}$

4.

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Braille letter

Simplifying an Expression, group *Gamma* γ -i

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $-5(9x + 5) - 4(-2 - x) \quad \dots \quad ??? \quad \dots \quad -41x - 17$

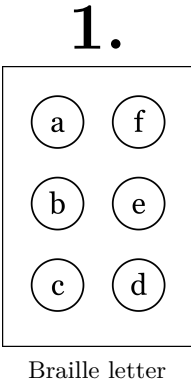
(b) $2(-2 - 2x)(-x - 1) + 4(4 + 2x) \quad ??? \quad 4x^2 - 16x$

(c) $(-x - 4)^3 - (-x - 5)^2 \quad ??? \quad x^3 + 13x^2 - 58x - 89$

(d) $\frac{5x - 2}{4} - 5 \cdot \frac{-1 + 3x}{7} \quad \dots \quad ??? \quad \dots \quad \frac{-25x + 6}{-28}$

(e) $\frac{\frac{2}{1} - \frac{-3}{x}}{\frac{1}{-5} + \frac{8}{1}} \quad \dots \quad ??? \quad \dots \quad \frac{-10x - 15}{-39x}$

(f) $\frac{(1 + 2x)^2 + 5}{(-x + 4) \cdot \frac{-1}{x}} \quad \dots \quad ??? \quad \dots \quad \frac{4x^3 + 4x^2 + 6x}{x - 4}$



Simplifying an Expression, group *Gamma* γ -ii

Name:

- (a) Expand the brackets and simplify the expression.

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(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $-2(-6x - 3) + 1(-1 - 3x) \quad \dots \quad ??? \quad \dots \quad 9x + 5$

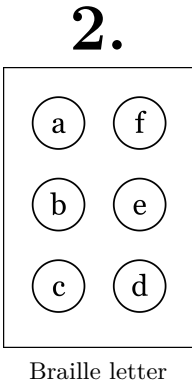
(b) $-3(2 - 2x)(2x - 3) - 4(-1 + 5x) \quad ??? \quad 12x^2 + 50x + 22$

(c) $(-4x - 4)^3 - (x + 3)^2 \quad ??? \quad -64x^3 - 193x^2 - 198x - 73$

(d) $\frac{6x - 3}{-2} - 4 \cdot \frac{-3 + 2x}{-2} \quad \dots \quad ??? \quad \dots \quad \frac{-18}{-4}$

(e) $\frac{\frac{7}{-3} - \frac{1}{x}}{\frac{1}{-2} + \frac{-5}{-1}} \quad \dots \quad ??? \quad \dots \quad \frac{14x + 6}{-27x}$

(f) $\frac{(-9 - x)^2 + 1}{(4x - 6) \cdot \frac{6}{x}} \quad \dots \quad ??? \quad \dots \quad \frac{x^3 + 18x^2 - 82x}{-24x - 36}$



Simplifying an Expression, group *Gamma* γ -iii

Name:

- (a) Expand the brackets and simplify the expression.

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(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $-9(-6x - 4) + 2(-1 + x) \quad \dots \quad ??? \quad \dots \quad 56x + 34$

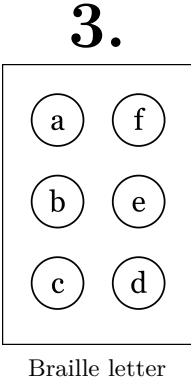
(b) $-2(-9 + 3x)(2x + 4) + 4(2 + 3x) \quad ??? \quad -12x^2 - 24x$

(c) $(-2x + 1)^3 - (x + 9)^2 \quad ??? \quad -8x^3 + 11x^2 - 24x - 80$

(d) $\frac{5x + 1}{2} + 2 \cdot \frac{-3 - 3x}{-7} \quad \dots \quad ??? \quad \dots \quad \frac{-47x - 19}{14}$

(e) $\frac{\frac{5}{9} - \frac{3}{x}}{\frac{1}{-1} + \frac{-2}{-3}} \quad \dots \quad ??? \quad \dots \quad \frac{15x - 81}{-9x}$

(f) $\frac{(-4 + 3x)^2 - 5}{(-2x - 6) \cdot \frac{3}{x}} \quad \dots \quad ??? \quad \dots \quad \frac{9x^3 - 24x^2 - 11x}{6x - 18}$



Simplifying an Expression, group *Gamma* γ -iv

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $5(-3x - 7) - 2(1 - x) \quad \dots \quad ??? \quad \dots \quad -13x - 37$

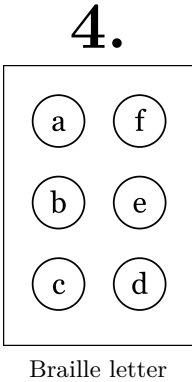
(b) $-5(-1 + 2x)(2x - 2) - 3(-6 + 6x) \quad ??? \quad -20x^2 + 12x + 8$

(c) $(2x - 3)^3 - (3x + 6)^2 \quad \dots \quad ??? \quad \dots \quad 8x^3 - 45x^2 + 18x - 63$

(d) $\frac{8x - 1}{-2} - 3 \cdot \frac{-4 + 4x}{3} \quad \dots \quad ??? \quad \dots \quad \frac{-48x - 27}{6}$

(e) $\frac{\frac{2}{-6} - \frac{-6}{x}}{\frac{1}{-1} + \frac{-5}{2}} \quad \dots \quad ??? \quad \dots \quad \frac{-4x + 72}{-42x}$

(f) $\frac{(4 - x)^2 + 2}{(x + 2) \cdot \frac{-1}{x}} \quad \dots \quad ??? \quad \dots \quad \frac{x^3 - 8x^2 - 18x}{x - 2}$



Simplifying an Expression, group *Delta* δ -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $4(-3x - 9) - 2(-6 - 6x) \dots\dots\dots ??? \dots\dots\dots -24$

(b) $2(3 + 2x)(-2x + 5) - 9(6 - 6x) \dots\dots\dots -8x^2 - 62x - 24$

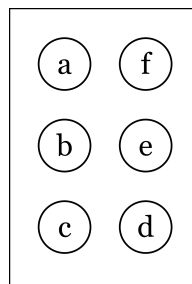
(c) $(2x - 1)^3 - (-6x - 5)^2 \dots\dots\dots -8x^3 - 48x^2 - 26$

(d) $\frac{-2x - 5}{-2} - 3 \cdot \frac{-6 - x}{4} \dots\dots\dots \frac{14x - 56}{8}$

(e) $\frac{\frac{1}{2} - \frac{6}{x}}{\frac{1}{1} + \frac{-3}{6}} \dots\dots\dots \frac{4x - 74}{6x}$

(f) $\frac{(3 + x)^2 - 2}{(3x - 6) \cdot \frac{7}{x}} \dots\dots\dots \frac{x^3 + 6x^2 - 7x}{-21x - 42}$

1.



Braille letter

Simplifying an Expression, group *Delta* δ -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-2(2x - 1) - 1(1 + 2x) \dots\dots\dots -6x - 1$

(b) $-7(-1 - 2x)(3x - 2) - 7(1 - 5x) \dots\dots\dots 42x^2 + 28x - 21$

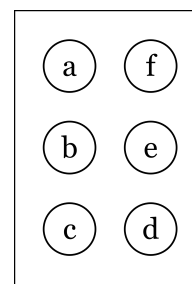
(c) $(-3x - 1)^3 - (-x + 2)^2 \dots\dots\dots -27x^3 - 28x^2 - 5x - 5$

(d) $\frac{-7x + 1}{-3} - 8 \cdot \frac{-1 - 4x}{-7} \dots\dots\dots \frac{-31}{-21}$

(e) $\frac{\frac{-4}{-2} - \frac{-5}{x}}{\frac{1}{-2} + \frac{-1}{-4}} \dots\dots\dots \frac{-32x - 80}{4x}$

(f) $\frac{(3 + x)^2 + 6}{(-8x + 1) \cdot \frac{-3}{x}} \dots\dots\dots \frac{x^3 + 6x^2 + 15x}{24x - 3}$

2.



Braille letter

Simplifying an Expression, group *Delta* δ -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-2(-2x + 1) - 1(4 + 5x) \dots\dots\dots -x - 6$

(b) $-4(5 - 2x)(4x + 1) + 8(7 + 9x) \dots\dots\dots 32x^2 + 2x$

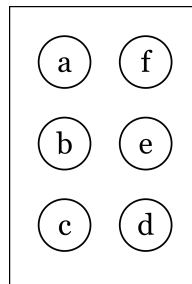
(c) $(-4x - 1)^3 - (-5x - 2)^2 \dots\dots\dots -64x^3 - 73x^2 - 32x - 5$

(d) $\frac{-4x - 2}{2} - 6 \cdot \frac{1 + 5x}{-4} \dots\dots\dots \frac{-44x - 4}{8}$

(e) $\frac{\frac{-6}{-1} - \frac{-2}{x}}{\frac{1}{1} + \frac{1}{8}} \dots\dots\dots \frac{-48x - 16}{-9x}$

(f) $\frac{(-2 - 3x)^2 + 1}{(-5x + 4) \cdot \frac{3}{x}} \dots\dots\dots \frac{9x^3 + 12x^2 - 5x}{15x + 12}$

3.



Braille letter

Simplifying an Expression, group *Delta* δ -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $8(x - 1) + 7(-4 + 3x) \dots\dots\dots 29x - 36$

(b) $2(-2 + 2x)(-9x - 6) - 1(-1 + 3x) \dots\dots\dots -36x^2 - 9x + 25$

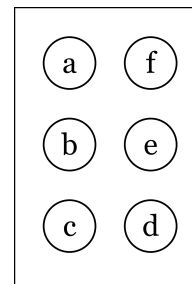
(c) $(4x + 4)^3 - (3x - 2)^2 \dots\dots\dots 64x^3 + 183x^2 + 204x + 60$

(d) $\frac{-2x + 4}{5} - 3 \cdot \frac{4 + 3x}{3} \dots\dots\dots \frac{-48}{-15}$

(e) $\frac{\frac{-3}{-2} - \frac{1}{x}}{\frac{1}{1} + \frac{6}{-1}} \dots\dots\dots \frac{6x - 3}{-10x}$

(f) $\frac{(6 + 6x)^2 - 4}{(-5x + 4) \cdot \frac{3}{x}} \dots\dots\dots \frac{36x^3 + 72x^2 + 32x}{-15x + 12}$

4.



Braille letter

Simplifying an Expression, group *Epsilon* € -i

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

(a) $-6(-x + 1) + 2(9 - 7x)$??? $-8x$

(b) $4(3 - 4x)(-5x - 4) + 4(-2 + 5x)$??? $80x^2 + 24x - 56$

(c) $(4x - 1)^3 - (x + 4)^2$... ??? ... $64x^3 - 49x^2 + 4x - 17$

(d) $\frac{3x - 9}{-4} - 3 \cdot \frac{-4 + 6x}{-2}$??? $\frac{-66x - 30}{-8}$

(e) $\frac{\frac{-9}{-5} - \frac{-2}{x}}{\frac{1}{3} + \frac{-3}{1}}$??? $\frac{-29x - 28}{40x}$

(f) $\frac{(-1 - 5x)^2 - 9}{(x - 7) \cdot \frac{-5}{x}}$??? $\frac{25x^3 + 10x^2 - 8x}{-5x + 35}$

1.

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Braille letter

Simplifying an Expression, group *Epsilon* € -ii

Name:

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- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

(a) $5(x + 3) + 9(4 + x)$??? $14x + 51$

(b) $4(2 - 5x)(-2x + 3) - 2(-2 - 5x)$??? $40x^2 + 66x$

(c) $(x - 1)^3 - (-8x + 2)^2$. ??? . $x^3 - 67x^2 + 35x - 5$

(d) $\frac{-4x - 5}{5} - 4 \cdot \frac{2 + 3x}{6}$??? $\frac{84x - 70}{-30}$

(e) $\frac{\frac{-7}{1} - \frac{6}{x}}{\frac{1}{-1} + \frac{-1}{1}}$??? $\frac{7x + 6}{2x}$

(f) $\frac{(-3 - x)^2 - 4}{(7x - 4) \cdot \frac{1}{x}}$??? $\frac{x^3 + 6x^2 - 5x}{-7x - 4}$

2.

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Braille letter

Simplifying an Expression, group *Epsilon* € -iii

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- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

(a) $7(7x + 1) - 1(-1 + x)$??? $48x + 8$

(b) $7(-2 - x)(4x - 1) - 9(8 + x)$??? $-28x^2 + 58x + 58$

(c) $(-x - 3)^3 - (3x - 3)^2$. ??? . $-x^3 - 18x^2 - 9x - 36$

(d) $\frac{-x - 1}{6} - 2 \cdot \frac{3 - 7x}{-3}$??? $\frac{87x - 33}{-18}$

(e) $\frac{\frac{8}{-8} - \frac{5}{x}}{\frac{1}{-1} + \frac{3}{-2}}$??? $\frac{19x + 77}{40x}$

(f) $\frac{(-7 - 5x)^2 + 5}{(x + 1) \cdot \frac{5}{x}}$??? $\frac{25x^3 + 70x^2 - 54x}{-5x + 5}$

3.

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Braille letter

Simplifying an Expression, group *Epsilon* € -iv

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- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

(a) $-3(4x - 3) + 6(3 - x)$??? $-18x + 27$

(b) $-5(1 - x)(-7x - 3) - 8(-1 + 4x)$??? $-35x^2 - 12x + 23$

(c) $(x - 4)^3 - (2x + 4)^2$??? $x^3 - 16x^2 + 32x - 80$

(d) $\frac{-8x - 5}{2} - 2 \cdot \frac{-4 + x}{5}$??? $\frac{44x - 9}{-10}$

(e) $\frac{\frac{-6}{3} - \frac{2}{x}}{\frac{1}{-2} + \frac{7}{6}}$??? $\frac{69x + 74}{-24x}$

(f) $\frac{(-8 - 8x)^2 + 6}{(2x - 8) \cdot \frac{-6}{x}}$??? $\frac{64x^3 + 128x^2 - 70x}{12x + 48}$

4.

a

f

b

e

c

d

Braille letter

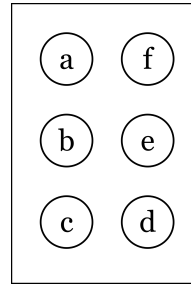
Simplifying an Expression, group *Zeta* ζ -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $5(-8x + 2) + 2(2 - x)$??? $-42x + 14$
 (b) $2(-2 + x)(4x - 7) + 1(-2 + x)$??? $8x^2 - 29x + 26$
 (c) $(2x + 4)^3 - (3x - 9)^2$... ??? ... $-8x^3 + 39x^2 - 17$
 (d) $\frac{-6x - 1}{-4} + 4 \cdot \frac{-1 + 3x}{-4}$??? $\frac{-24x + 20}{-16}$
 (e) $\frac{\frac{-6}{-6} - \frac{-6}{x}}{\frac{1}{1} + \frac{7}{1}}$??? $\frac{-5x - 34}{-48x}$
 (f) $\frac{(6 + 8x)^2 - 4}{(-2x - 9) \cdot \frac{-6}{x}}$??? $\frac{64x^3 + 96x^2 + 32x}{12x + 54}$

1.



Braille letter

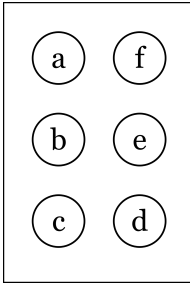
Simplifying an Expression, group *Zeta* ζ -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $2(4x - 9) + 1(3 - 3x)$??? $5x$
 (b) $-3(2 + x)(2x + 2) - 3(-2 + 2x)$??? $-6x^2 - 24x - 6$
 (c) $(-2x + 3)^3 - (-2x - 4)^2$??? $8x^3 + 32x^2 - 70x + 11$
 (d) $\frac{2x - 2}{-6} - 2 \cdot \frac{-2 - 2x}{-2}$??? $\frac{-28x - 20}{-12}$
 (e) $\frac{\frac{7}{6} - \frac{-4}{x}}{\frac{1}{1} + \frac{-8}{1}}$??? $\frac{4x + 25}{-42x}$
 (f) $\frac{(2 - 8x)^2 - 2}{(-9x + 6) \cdot \frac{-8}{x}}$??? $\frac{64x^3 - 32x^2 + 2x}{72x - 48}$

2.



Braille letter

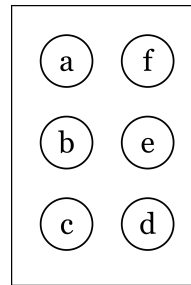
Simplifying an Expression, group *Zeta* ζ -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $-3(-x - 2) - 5(-7 - 3x)$??? $18x + 41$
 (b) $-3(-1 + 2x)(x - 2) + 3(-7 - 2x)$??? $-6x^2 + 9x - 27$
 (c) $(2x + 1)^3 - (-2x - 3)^2$... ??? ... $8x^3 + 8x^2 - 6x - 8$
 (d) $\frac{-5x + 1}{-6} + 3 \cdot \frac{2 + 3x}{2}$??? $\frac{-64x - 34}{-12}$
 (e) $\frac{\frac{4}{1} - \frac{-1}{x}}{\frac{1}{-8} + \frac{2}{-1}}$??? $\frac{35x + 7}{-17x}$
 (f) $\frac{(1 - 3x)^2 - 3}{(-x - 2) \cdot \frac{9}{x}}$??? $\frac{9x^3 - 6x^2 + 2x}{9x - 18}$

3.



Braille letter

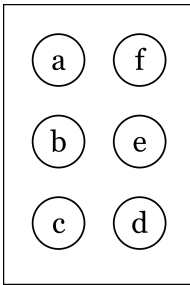
Simplifying an Expression, group *Zeta* ζ -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $-6(-2x + 1) - 5(-2 - 3x)$??? $27x + 4$
 (b) $-6(2 + 2x)(3x + 4) + 8(-4 + 2x)$??? $-36x^2 + 68x - 80$
 (c) $(-3x - 1)^3 - (-7x - 5)^2$. ??? . $27x^3 + 76x^2 - 79x - 26$
 (d) $\frac{6x + 5}{3} + 4 \cdot \frac{1 - x}{6}$??? $\frac{42}{-18}$
 (e) $\frac{\frac{2}{1} - \frac{8}{x}}{\frac{1}{6} + \frac{4}{1}}$??? $\frac{12x - 48}{25x}$
 (f) $\frac{(7 - 3x)^2 + 1}{(3x + 7) \cdot \frac{-7}{x}}$??? $\frac{9x^3 - 42x^2 - 50x}{21x - 49}$

4.



Braille letter

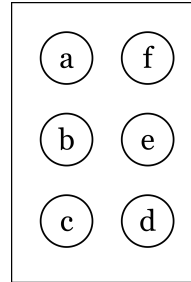
Simplifying an Expression, group *Eta η* -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $3(-3x - 1) - 6(1 - 5x)$??? $21x - 9$
 (b) $-8(-1 - 5x)(2x - 2) + 7(-3 + 3x)$??? $80x^2 - 43x - 37$
 (c) $(-3x + 4)^3 - (-2x + 3)^2$??? $27x^3 - 104x^2 - 132x + 55$
 (d) $\frac{-3x + 1}{-2} - 7 \cdot \frac{-1 - 2x}{-9}$??? $\frac{-x - 23}{-18}$
 (e) $\frac{\frac{-2}{-7} - \frac{-1}{x}}{\frac{1}{-1} + \frac{2}{1}}$??? $\frac{2x + 7}{7x}$
 (f) $\frac{(-3 - x)^2 - 1}{(-3x - 2) \cdot \frac{-4}{x}}$??? $\frac{x^3 + 6x^2 + 8x}{12x + 8}$

1.



Braille letter

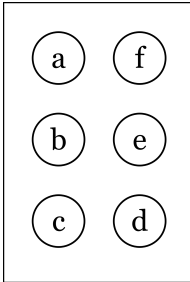
Simplifying an Expression, group *Eta η* -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $-7(2x + 2) - 3(-5 - 3x)$??? $-5x + 1$
 (b) $-2(6 + 3x)(-5x + 2) - 1(-3 + 3x)$??? $30x^2 - 45x$
 (c) $(-2x - 1)^3 - (4x + 4)^2$??? $8x^3 + 28x^2 - 38x - 17$
 (d) $\frac{-x - 9}{6} - 5 \cdot \frac{1 - x}{-6}$??? $\frac{-36x + 24}{36}$
 (e) $\frac{\frac{-4}{-1} - \frac{-4}{x}}{\frac{1}{-4} + \frac{2}{-1}}$??? $\frac{-14x - 18}{9x}$
 (f) $\frac{(3 + 5x)^2 + 6}{(-5x - 4) \cdot \frac{3}{x}}$??? $\frac{25x^3 + 30x^2 - 15x}{15x - 12}$

2.



Braille letter

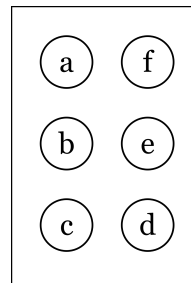
Simplifying an Expression, group *Eta η* -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $4(7x + 2) - 8(5 + 4x)$??? $-4x - 32$
 (b) $3(-2 + 4x)(-5x + 4) + 3(-7 + 3x)$??? $-60x^2 - 87x$
 (c) $(x + 1)^3 - (2x + 1)^2$??? $x^3 - x^2 - x$
 (d) $\frac{-2x + 6}{-2} - 5 \cdot \frac{5 + 2x}{4}$??? $\frac{12x + 74}{8}$
 (e) $\frac{\frac{-7}{-6} - \frac{-6}{x}}{\frac{1}{1} + \frac{5}{-1}}$??? $\frac{8x + 34}{-24x}$
 (f) $\frac{(-6 + x)^2 - 6}{(2x - 6) \cdot \frac{-1}{x}}$??? $\frac{x^3 - 12x^2 + 30x}{-2x + 6}$

3.



Braille letter

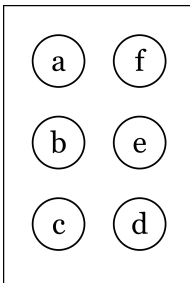
Simplifying an Expression, group *Eta η* -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $3(-x + 7) - 2(-2 + x)$??? $-5x + 25$
 (b) $-2(-4 + x)(-x + 3) - 3(1 - 3x)$??? $2x^2 + 5x - 21$
 (c) $(-4x + 2)^3 - (6x + 4)^2$... ??? ... $64x^3 + 60x^2 - 8$
 (d) $\frac{-4x - 8}{-8} - 3 \cdot \frac{4 + x}{6}$??? $\frac{48}{48}$
 (e) $\frac{\frac{-6}{3} - \frac{6}{x}}{\frac{1}{-1} + \frac{-5}{3}}$??? $\frac{18x + 54}{24x}$
 (f) $\frac{(-5 - 2x)^2 + 6}{(2x - 7) \cdot \frac{-5}{x}}$??? $\frac{4x^3 + 20x^2 - 31x}{10x + 35}$

4.



Braille letter

Simplifying an Expression, group *Theta* θ -i

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $-2(-2x + 5) + 2(3 - 2x)$??? -4

(b) $5(2 - 6x)(-3x - 3) + 4(1 - 4x)$??? $90x^2 + 44x - 26$

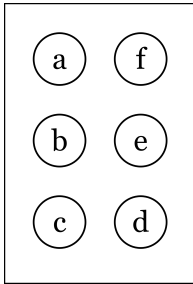
(c) $(-x + 3)^3 - (-9x - 6)^2$. ??? . $x^3 + 72x^2 - 135x - 9$

(d) $\frac{-2x + 6}{-3} - 5 \cdot \frac{2 + 4x}{3}$??? $\frac{54x + 48}{9}$

(e) $\frac{\frac{-2}{1} - \frac{-5}{x}}{\frac{1}{2} + \frac{-3}{1}}$??? $\frac{-4x + 10}{-5x}$

(f) $\frac{(4 - 6x)^2 - 5}{(5x - 3) \cdot \frac{-2}{x}}$??? $\frac{36x^3 - 48x^2 + 11x}{-10x + 6}$

1.



Braille letter

Simplifying an Expression, group *Theta* θ -ii

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $6(x + 1) - 9(-2 + 5x)$??? $-39x + 24$

(b) $-2(-3 + 4x)(2x + 4) + 8(-1 + x)$??? $-16x^2 + 12x + 16$

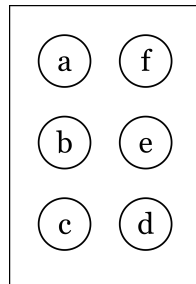
(c) $(-2x - 3)^3 - (-8x - 8)^2$??? $-8x^3 - 100x^2 - 182x - 91$

(d) $\frac{-7x - 9}{8} - 3 \cdot \frac{-1 + 3x}{4}$??? $\frac{-100x - 12}{-32}$

(e) $\frac{\frac{-2}{-2} - \frac{7}{x}}{\frac{1}{3} + \frac{5}{-2}}$??? $\frac{12x - 84}{-26x}$

(f) $\frac{(1 - x)^2 + 4}{(8x + 5) \cdot \frac{-3}{x}}$??? $\frac{x^3 - 2x^2 - 5x}{24x - 15}$

2.



Braille letter

Simplifying an Expression, group *Theta* θ -iii

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $3(x - 8) - 1(4 + 8x)$??? $-5x - 28$

(b) $-4(-1 + 2x)(6x + 3) + 6(8 + 4x)$??? $-48x^2 + 24x + 60$

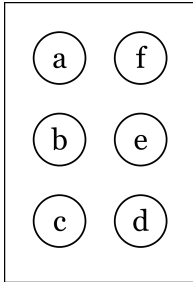
(c) $(-4x - 2)^3 - (-2x - 4)^2$??? $-64x^3 - 100x^2 - 64x - 24$

(d) $\frac{x - 1}{2} + 3 \cdot \frac{1 - 5x}{-3}$??? $\frac{33x + 9}{6}$

(e) $\frac{\frac{-3}{-1} - \frac{1}{x}}{\frac{1}{-2} + \frac{3}{-1}}$??? $\frac{-5x - 1}{7x}$

(f) $\frac{(1 - 3x)^2 + 2}{(-x - 7) \cdot \frac{-9}{x}}$??? $\frac{9x^3 - 6x^2 - 3x}{-9x + 63}$

3.



Braille letter

Simplifying an Expression, group *Theta* θ -iv

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $-2(3x - 1) - 6(-5 - 2x)$??? $6x + 32$

(b) $2(2 + 6x)(7x - 4) - 3(9 + 3x)$??? $84x^2 + 29x - 43$

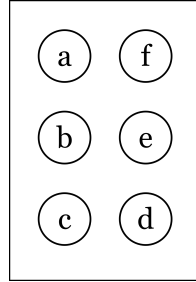
(c) $(3x - 3)^3 - (3x - 3)^2$??? $-27x^3 + 90x^2 + 99x - 36$

(d) $\frac{-4x - 4}{-6} + 2 \cdot \frac{1 + 5x}{-7}$??? $\frac{16}{-42}$

(e) $\frac{\frac{-2}{3} - \frac{5}{x}}{\frac{1}{1} + \frac{1}{5}}$??? $\frac{-10x - 75}{18x}$

(f) $\frac{(-5 - x)^2 + 2}{(2x - 3) \cdot \frac{1}{x}}$??? $\frac{x^3 + 10x^2 + 27x}{2x - 3}$

4.



Braille letter

Simplifying an Expression, group *Iota ι* -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-2(3x - 4) + 4(-4 + 6x)$??? $18x - 8$

(b) $3(3 - 5x)(-4x + 1) - 8(-5 + 7x)$??? $60x^2 - 107x + 49$

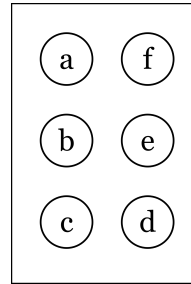
(c) $(-2x + 2)^3 - (5x + 2)^2$... ??? ... $-8x^3 - x^2 - 44x + 4$

(d) $\frac{-4x - 4}{3} - 4 \cdot \frac{-4 + 5x}{-2}$??? $\frac{56}{6}$

(e) $\frac{\frac{1}{2} - \frac{-4}{x}}{\frac{1}{-8} + \frac{-2}{-1}}$??? $\frac{5x + 35}{15x}$

(f) $\frac{(-6 - 4x)^2 - 6}{(-4x - 1) \cdot \frac{2}{x}}$??? $\frac{16x^3 + 48x^2 - 30x}{8x - 2}$

1.



Braille letter

Simplifying an Expression, group *Iota ι* -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-5(4x - 2) - 4(-5 + 4x)$??? $-36x - 30$

(b) $-2(-1 + 3x)(4x + 3) - 3(-2 - 7x)$??? $-24x^2 + 11x + 12$

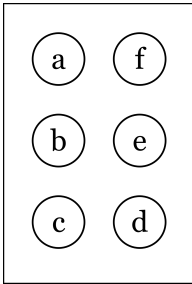
(c) $(x - 3)^3 - (2x - 7)^2$??? $-x^3 + 13x^2 + 55x - 76$

(d) $\frac{-x + 3}{-7} - 3 \cdot \frac{3 + 2x}{6}$??? $\frac{81}{42}$

(e) $\frac{\frac{-4}{2} - \frac{-1}{x}}{\frac{1}{1} + \frac{-1}{-5}}$??? $\frac{22x - 7}{-12x}$

(f) $\frac{(7 - 6x)^2 + 4}{(-8x + 3) \cdot \frac{7}{x}}$??? $\frac{36x^3 - 84x^2 + 53x}{-56x + 21}$

2.



Braille letter

Simplifying an Expression, group *Iota ι* -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $2(-5x - 7) - 2(6 - 4x)$??? $-2x - 26$

(b) $2(3 - 2x)(x - 1) - 2(-3 - 5x)$.. ??? .. $-4x^2 - 20x$

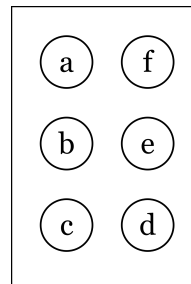
(c) $(-2x - 4)^3 - (5x - 2)^2$??? $-8x^3 - 73x^2 - 76x - 68$

(d) $\frac{-x + 7}{-4} - 6 \cdot \frac{-1 - 4x}{-9}$??? $\frac{-87}{-36}$

(e) $\frac{\frac{4}{2} - \frac{-7}{x}}{\frac{1}{1} + \frac{-3}{-6}}$??? $\frac{-24x - 84}{-18x}$

(f) $\frac{(1 + 2x)^2 - 8}{(-x + 4) \cdot \frac{-2}{x}}$??? $\frac{4x^3 + 4x^2 - 7x}{2x - 8}$

3.



Braille letter

Simplifying an Expression, group *Iota ι* -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $9(x + 1) + 3(4 - 7x)$??? $-12x + 21$

(b) $3(-5 + 3x)(x - 4) + 6(8 + 4x)$??? $9x^2 + 27x - 108$

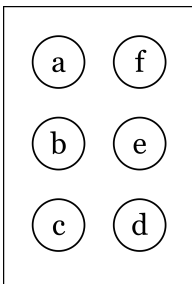
(c) $(3x + 2)^3 - (4x - 6)^2$... ??? ... $-27x^3 - 38x^2 - 28$

(d) $\frac{2x - 7}{2} + 4 \cdot \frac{1 - x}{8}$??? $\frac{-48}{-16}$

(e) $\frac{\frac{-1}{1} - \frac{4}{x}}{\frac{1}{8} + \frac{8}{-2}}$??? $\frac{16x + 64}{62x}$

(f) $\frac{(4 - 4x)^2 + 1}{(-7x - 2) \cdot \frac{-3}{x}}$??? $\frac{16x^3 - 32x^2 - 17x}{-21x + 6}$

4.



Braille letter

Simplifying an Expression, group *Kappa* κ -i

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

- (a) $-2(-7x + 6) + 5(-7 + x)$??? $19x - 47$
- (b) $-3(3 - 3x)(2x - 3) + 2(6 + 9x)$??? $18x^2 - 27x + 39$
- (c) $(4x + 1)^3 - (-5x + 3)^2$. ??? . $64x^3 + 23x^2 + 42x - 8$
- (d) $\frac{-8x - 7}{-3} + 6 \cdot \frac{-4 - 3x}{-7}$??? $\frac{110x + 121}{21}$
- (e) $\frac{\frac{2}{-1} - \frac{1}{x}}{\frac{1}{-6} + \frac{8}{-2}}$??? $\frac{24x + 12}{50x}$
- (f) $\frac{(2 - 3x)^2 + 8}{(-x - 6) \cdot \frac{2}{x}}$??? $\frac{9x^3 - 12x^2 - 12x}{2x - 12}$

1.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Kappa* κ -ii

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

- (a) $4(5x + 3) - 6(2 - 5x)$??? $50x$
- (b) $-8(4 + 2x)(2x + 2) - 2(1 + 9x)$??? $-32x^2 - 114x - 66$
- (c) $(-4x + 4)^3 - (x + 5)^2$. ??? . $64x^3 - 191x^2 - 202x + 39$
- (d) $\frac{9x - 8}{-2} + 5 \cdot \frac{3 + 8x}{4}$??? $\frac{-62}{8}$
- (e) $\frac{\frac{6}{-1} - \frac{4}{x}}{\frac{1}{3} + \frac{4}{2}}$??? $\frac{35x + 22}{-14x}$
- (f) $\frac{(-6 + 2x)^2 + 3}{(-3x - 6) \cdot \frac{1}{x}}$??? $\frac{4x^3 - 24x^2 + 39x}{-3x - 6}$

2.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Kappa* κ -iii

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

- (a) $3(-3x + 8) - 1(1 - x)$??? $-8x + 23$
- (b) $-2(3 - 7x)(-x - 4) + 2(-1 + 5x)$??? $-14x^2 + 40x + 22$
- (c) $(2x - 1)^3 - (-8x + 2)^2$... ??? ... $8x^3 - 76x^2 + 38x - 5$
- (d) $\frac{-5x + 4}{5} - 3 \cdot \frac{6 - 2x}{2}$??? $\frac{-82}{-10}$
- (e) $\frac{\frac{2}{-5} - \frac{-2}{x}}{\frac{1}{3} + \frac{2}{1}}$??? $\frac{6x - 30}{-35x}$
- (f) $\frac{(7 - 6x)^2 - 2}{(-4x + 5) \cdot \frac{2}{x}}$??? $\frac{36x^3 - 84x^2 + 47x}{-8x + 10}$

3.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Kappa* κ -iv

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

- (a) $8(8x + 2) - 3(-1 + x)$??? $61x + 19$
- (b) $-3(-3 + 4x)(x + 1) - 3(-8 - 8x)$??? $-12x^2 - 21x - 33$
- (c) $(-x + 2)^3 - (7x - 1)^2$??? $x^3 + 43x^2 + 7$
- (d) $\frac{-4x - 6}{-7} - 2 \cdot \frac{1 + 5x}{-6}$??? $\frac{-94x + 50}{-42}$
- (e) $\frac{\frac{-2}{2} - \frac{-3}{x}}{\frac{1}{4} + \frac{-8}{-4}}$??? $\frac{32x - 96}{-72x}$
- (f) $\frac{(1 + x)^2 - 1}{(x + 9) \cdot \frac{-1}{x}}$??? $\frac{x^3 + 2x^2}{-x - 9}$

4.

a

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d

Braille letter

Simplifying an Expression, group *Lambda* λ -i

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $6(9x + 6) - 1(-9 - 8x)$??? $62x + 45$

(b) $-6(-1 - 3x)(-2x + 1) - 8(-6 - 8x)$??? $-36x^2 - 70x$

(c) $(4x - 1)^3 - (8x - 5)^2$. ??? . $64x^3 - 112x^2 + 92x - 26$

(d) $\frac{2x + 4}{-3} + 9 \cdot \frac{-1 - x}{8}$??? $\frac{43x + 59}{-24}$

(e) $\frac{\frac{1}{-5} - \frac{-1}{x}}{\frac{1}{-1} + \frac{4}{-4}}$??? $\frac{6x - 21}{40x}$

(f) $\frac{(-6 + 4x)^2 - 4}{(-x + 3) \cdot \frac{1}{x}}$??? $\frac{16x^3 - 48x^2 - 32x}{x + 3}$

1.

a

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Braille letter

Simplifying an Expression, group *Lambda* λ -ii

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $-3(-4x - 4) + 3(1 - 2x)$??? $6x$

(b) $-2(-1 - 3x)(2x + 3) + 3(1 - x)$??? $12x^2 + 19x + 9$

(c) $(x + 4)^3 - (-x - 1)^2$.. ??? .. $x^3 + 11x^2 + 46x + 63$

(d) $\frac{5x - 2}{-5} - 2 \cdot \frac{4 - x}{6}$??? $\frac{-20x + 28}{30}$

(e) $\frac{\frac{-6}{-9} - \frac{2}{x}}{\frac{1}{-1} + \frac{9}{-1}}$??? $\frac{-9x + 17}{90x}$

(f) $\frac{(5 + 7x)^2 + 2}{(x - 7) \cdot \frac{1}{x}}$??? $\frac{49x^3 + 70x^2 + 27x}{x - 7}$

2.

a

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Braille letter

Simplifying an Expression, group *Lambda* λ -iii

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $3(-x + 6) - 1(6 + 2x)$??? $-5x + 12$

(b) $5(6 + x)(-2x - 2) + 1(1 - 3x)$??? $-10x^2 + 73x + 59$

(c) $(-4x + 1)^3 - (-x + 8)^2$... ??? ... $64x^3 - 47x^2 - 63$

(d) $\frac{-3x + 2}{5} + 2 \cdot \frac{-1 - 8x}{-2}$??? $\frac{-14}{10}$

(e) $\frac{\frac{-3}{4} - \frac{6}{x}}{\frac{1}{-1} + \frac{2}{1}}$??? $\frac{3x + 24}{-4x}$

(f) $\frac{(-5 + 5x)^2 + 4}{(8x + 1) \cdot \frac{6}{x}}$??? $\frac{25x^3 - 50x^2 - 29x}{-48x + 6}$

3.

a

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Braille letter

Simplifying an Expression, group *Lambda* λ -iv

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $8(-5x + 2) + 2(2 + 2x)$??? $-36x + 20$

(b) $-4(-1 + x)(3x + 7) - 4(2 + 2x)$??? $-12x^2 - 24x + 20$

(c) $(-2x - 1)^3 - (x + 6)^2$. ??? . $-8x^3 - 13x^2 - 18x - 37$

(d) $\frac{-5x - 5}{-2} + 4 \cdot \frac{2 - x}{-5}$??? $\frac{33x + 9}{-10}$

(e) $\frac{\frac{1}{-4} - \frac{4}{x}}{\frac{1}{-1} + \frac{-9}{6}}$??? $\frac{-6x - 96}{-60x}$

(f) $\frac{(-4 - 3x)^2 + 5}{(7x + 4) \cdot \frac{-5}{x}}$??? $\frac{9x^3 + 24x^2 - 21x}{35x - 20}$

4.

a

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d

Braille letter

Simplifying an Expression, group *Mu* μ -i

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $-6(-x + 5) + 7(-2 - 2x)$??? $-8x - 44$

(b) $4(4 + x)(3x - 3) + 5(1 + 3x)$??? $12x^2 + 51x - 43$

(c) $(2x + 4)^3 - (x + 2)^2$??? $-8x^3 + 47x^2 + 92x + 60$

(d) $\frac{5x - 6}{-4} - 2 \cdot \frac{-2 + 3x}{8}$??? $\frac{-64x - 64}{32}$

(e) $\frac{\frac{-1}{-4} - \frac{-4}{x}}{\frac{1}{3} + \frac{-3}{1}}$??? $\frac{-4x - 49}{32x}$

(f) $\frac{(2 - x)^2 - 6}{(2x + 2) \cdot \frac{3}{x}}$??? $\frac{x^3 - 4x^2 + 2x}{-6x + 6}$

1.

a

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Braille letter

Simplifying an Expression, group *Mu* μ -ii

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $7(-x - 2) + 5(7 - 3x)$??? $-22x + 21$

(b) $2(5 + 2x)(7x - 4) + 2(-5 + 2x)$??? $28x^2 - 58x + 50$

(c) $(x - 3)^3 - (-6x - 1)^2$.. ??? .. $x^3 - 45x^2 + 15x - 28$

(d) $\frac{-x + 5}{-3} + 2 \cdot \frac{-5 - 2x}{-2}$??? $\frac{20}{-6}$

(e) $\frac{\frac{5}{-2} - \frac{4}{x}}{\frac{1}{-3} + \frac{4}{-1}}$??? $\frac{15x + 24}{26x}$

(f) $\frac{(-1 + 3x)^2 + 1}{(-x - 2) \cdot \frac{3}{x}}$??? $\frac{9x^3 - 6x^2 - 2x}{3x - 6}$

2.

a

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b

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d

Braille letter

Simplifying an Expression, group *Mu* μ -iii

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $-6(-7x + 2) - 3(-8 + 5x)$??? $27x$

(b) $5(-4 + x)(5x + 3) + 6(3 + 4x)$??? $25x^2 - 61x - 42$

(c) $(-3x - 1)^3 - (-x + 2)^2$??? $-27x^3 - 28x^2 - 5x - 5$

(d) $\frac{-2x - 3}{3} + 2 \cdot \frac{-3 + 3x}{5}$??? $\frac{-8x - 33}{-15}$

(e) $\frac{\frac{1}{-1} - \frac{2}{x}}{\frac{1}{2} + \frac{-5}{-2}}$??? $\frac{-6x - 10}{12x}$

(f) $\frac{(-1 + 4x)^2 + 3}{(-x - 8) \cdot \frac{6}{x}}$??? $\frac{16x^3 - 8x^2 + 4x}{-6x - 48}$

3.

a

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d

Braille letter

Simplifying an Expression, group *Mu* μ -iv

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $5(3x - 6) + 5(7 - 7x)$??? $-20x$

(b) $4(-4 + 3x)(-4x + 2) + 1(-6 + x)$??? $-48x^2 + 89x - 38$

(c) $(-4x + 2)^3 - (8x + 5)^2$. ??? . $-64x^3 + 32x^2 - 128x - 17$

(d) $\frac{-6x - 1}{4} + 7 \cdot \frac{2 - x}{6}$??? $\frac{64x + 50}{-24}$

(e) $\frac{\frac{2}{1} - \frac{-1}{x}}{\frac{1}{-6} + \frac{2}{-1}}$??? $\frac{9x + 4}{-13x}$

(f) $\frac{(-1 + x)^2 + 4}{(3x + 5) \cdot \frac{1}{x}}$??? $\frac{x^3 - 2x^2 + 5x}{3x + 5}$

4.

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Braille letter

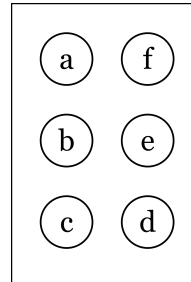
Simplifying an Expression, group $Nu \nu$ -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $7(-6x - 1) + 3(-2 - 9x) \dots \dots \dots ??? \dots \dots -69x - 13$
 (b) $2(2 + 3x)(2x + 7) + 3(7 - 3x) \dots \dots \dots 12x^2 + 41x + 49$
 (c) $(-x - 1)^3 - (-3x + 3)^2 \dots \dots \dots x^3 - 12x^2 + 15x - 10$
 (d) $\frac{x + 4}{-2} - 5 \cdot \frac{-3 - 4x}{-8} \dots \dots \dots ??? \dots \dots \frac{48x - 62}{-16}$
 (e) $\frac{\frac{-7}{2} - \frac{6}{x}}{\frac{1}{1} + \frac{-6}{4}} \dots \dots \dots ??? \dots \dots \frac{-28x - 48}{-4x}$
 (f) $\frac{(1 - x)^2 - 9}{(-9x + 5) \cdot \frac{4}{x}} \dots \dots \dots ??? \dots \dots \frac{x^3 - 2x^2 - 8x}{-36x + 20}$

1.



Braille letter

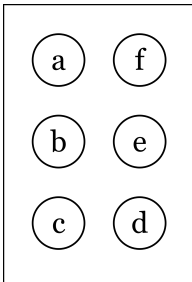
Simplifying an Expression, group $Nu \nu$ -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $4(4x - 7) + 6(-8 - 8x) \dots \dots \dots ??? \dots \dots -32x - 76$
 (b) $-3(-3 + 4x)(5x - 4) + 9(-4 + 6x) \dots \dots \dots -60x^2 - 147x - 72$
 (c) $(3x + 2)^3 - (-8x + 1)^2 \dots \dots \dots 27x^3 - 10x^2 + 52x + 7$
 (d) $\frac{-5x - 2}{-7} - 2 \cdot \frac{1 - 2x}{-7} \dots \dots \dots ??? \dots \dots \frac{-7x + 28}{-49}$
 (e) $\frac{\frac{-4}{1} - \frac{-5}{x}}{\frac{1}{-5} + \frac{8}{3}} \dots \dots \dots ??? \dots \dots \frac{60x - 75}{-37x}$
 (f) $\frac{(5 + 2x)^2 + 3}{(-2x + 1) \cdot \frac{-4}{x}} \dots \dots \dots ??? \dots \dots \frac{4x^3 + 20x^2 - 28x}{-8x - 4}$

2.



Braille letter

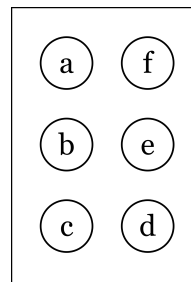
Simplifying an Expression, group $Nu \nu$ -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $7(-4x + 2) - 6(2 + 5x) \dots \dots \dots ??? \dots \dots -58x + 2$
 (b) $-6(-5 - x)(-x + 1) + 7(5 + 4x) \dots \dots \dots -6x^2 - 4x + 65$
 (c) $(4x + 1)^3 - (6x - 2)^2 \dots \dots \dots -64x^3 + 12x^2 + 36x - 3$
 (d) $\frac{-2x - 6}{2} + 4 \cdot \frac{-4 + x}{7} \dots \dots \dots ??? \dots \dots \frac{6x - 74}{-14}$
 (e) $\frac{\frac{-2}{-2} - \frac{6}{x}}{\frac{1}{-3} + \frac{6}{-1}} \dots \dots \dots ??? \dots \dots \frac{-4x + 35}{38x}$
 (f) $\frac{(-5 + x)^2 + 2}{(x + 7) \cdot \frac{7}{x}} \dots \dots \dots ??? \dots \dots \frac{x^3 - 10x^2 - 27x}{-7x + 49}$

3.



Braille letter

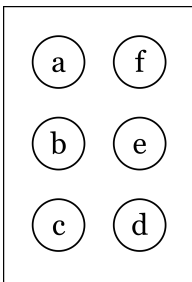
Simplifying an Expression, group $Nu \nu$ -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $-6(-2x + 1) - 2(2 + 4x) \dots \dots \dots ??? \dots \dots 4x - 10$
 (b) $4(5 + x)(-2x + 3) - 4(-6 - 2x) \dots \dots \dots -8x^2 - 20x + 84$
 (c) $(-x + 2)^3 - (x - 6)^2 \dots \dots \dots -x^3 + 5x^2 - 28$
 (d) $\frac{-5x - 9}{-7} - 2 \cdot \frac{-3 + 6x}{4} \dots \dots \dots ??? \dots \dots \frac{-64x - 78}{28}$
 (e) $\frac{\frac{-5}{4} - \frac{-2}{x}}{\frac{1}{-9} + \frac{1}{-1}} \dots \dots \dots ??? \dots \dots \frac{-47x + 71}{-40x}$
 (f) $\frac{(1 - 7x)^2 + 6}{(3x - 3) \cdot \frac{-4}{x}} \dots \dots \dots ??? \dots \dots \frac{49x^3 - 14x^2 - 7x}{12x + 12}$

4.



Braille letter

Simplifying an Expression, group $Xi \xi$ -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-2(3x + 5) - 4(-4 - 2x)$??? $2x - 6$

(b) $7(-3 - x)(-2x + 3) - 2(-1 + 6x)$??? $14x^2 + 9x - 61$

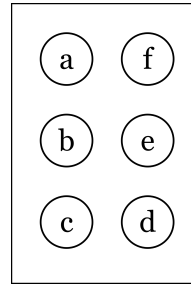
(c) $(-x + 2)^3 - (-6x - 1)^2$. ??? . $-x^3 - 30x^2 - 24x + 7$

(d) $\frac{-x-1}{-4} + 4 \cdot \frac{-4-6x}{-3}$??? $\frac{67}{-12}$

(e) $\frac{\frac{-2}{2} - \frac{-2}{x}}{\frac{1}{-2} + \frac{5}{-3}}$??? $\frac{-14x + 26}{-26x}$

(f) $\frac{(2-3x)^2 - 3}{(4x-2) \cdot \frac{-2}{x}}$??? $\frac{9x^3 - 12x^2 + x}{-8x + 4}$

1.



Braille letter

Simplifying an Expression, group $Xi \xi$ -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $4(-4x + 2) + 4(6 - 5x)$??? $-36x + 32$

(b) $5(3 + 5x)(-4x - 2) + 8(2 + 6x)$??? $-100x^2 + 62x - 14$

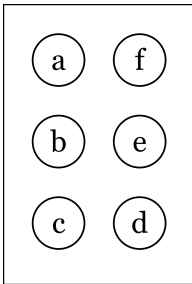
(c) $(2x - 1)^3 - (-7x + 4)^2$. ??? . $-8x^3 - 61x^2 + 62x - 17$

(d) $\frac{-x-4}{2} - 3 \cdot \frac{-3-6x}{6}$??? $\frac{30x-6}{-12}$

(e) $\frac{\frac{1}{-1} - \frac{3}{x}}{\frac{1}{-6} + \frac{-6}{2}}$??? $\frac{-10x-39}{-38x}$

(f) $\frac{(1+6x)^2 + 5}{(6x-1) \cdot \frac{4}{x}}$??? $\frac{36x^3 + 12x^2 - 6x}{-24x - 4}$

2.



Braille letter

Simplifying an Expression, group $Xi \xi$ -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $3(5x + 3) + 3(-5 - x)$??? $12x - 6$

(b) $-5(3 - 4x)(-2x + 1) + 1(6 + 2x)$??? $-40x^2 + 52x - 9$

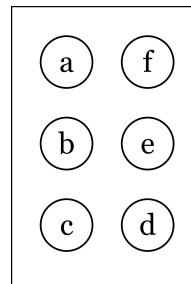
(c) $(-2x - 1)^3 - (-7x - 4)^2$??? $-8x^3 - 61x^2 - 62x - 17$

(d) $\frac{-2x+2}{4} + 3 \cdot \frac{2+8x}{8}$??? $\frac{80x+40}{32}$

(e) $\frac{\frac{-3}{2} - \frac{2}{x}}{\frac{1}{6} + \frac{2}{1}}$??? $\frac{-21x-25}{26x}$

(f) $\frac{(-9-7x)^2 + 4}{(2x+5) \cdot \frac{-1}{x}}$??? $\frac{49x^3 + 126x^2 - 85x}{2x-5}$

3.



Braille letter

Simplifying an Expression, group $Xi \xi$ -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-5(-x + 2) + 2(-4 + 2x)$??? $9x - 18$

(b) $-8(3 + 3x)(-x - 4) - 4(-4 - 4x)$??? $24x^2 - 136x - 112$

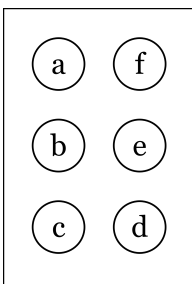
(c) $(3x + 1)^3 - (8x - 6)^2$??? $-27x^3 - 37x^2 - 35$

(d) $\frac{-9x+7}{-3} + 5 \cdot \frac{-1-2x}{3}$??? $\frac{36}{9}$

(e) $\frac{\frac{-1}{-2} - \frac{-4}{x}}{\frac{1}{1} + \frac{1}{4}}$??? $\frac{-4x-32}{-10x}$

(f) $\frac{(1+2x)^2 + 8}{(-2x+3) \cdot \frac{3}{x}}$??? $\frac{4x^3 + 4x^2 - 9x}{6x+9}$

4.



Braille letter

Simplifying an Expression, group *Omicron o* -i

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

(a) $9(2x - 9) + 4(3 + x)$??? $22x$

(b) $3(-2 - 3x)(2x - 4) - 4(-3 + 7x)$??? $-18x^2 - 4x + 36$

(c) $(2x - 3)^3 - (7x - 3)^2$... ??? ... $8x^3 - 85x^2 + 96x - 36$

(d) $\frac{-5x + 3}{-3} - 5 \cdot \frac{3 + 5x}{-4}$??? $\frac{33}{-12}$

(e) $\frac{\frac{-5}{1} - \frac{-1}{x}}{\frac{1}{-5} + \frac{1}{-1}}$??? $\frac{-22x + 4}{-6x}$

(f) $\frac{(1 - 2x)^2 - 1}{(-2x - 7) \cdot \frac{2}{x}}$??? $\frac{4x^3 - 4x^2}{-4x - 14}$

1.

a

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b

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d

Braille letter

Simplifying an Expression, group *Omicron o* -ii

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

(a) $2(-3x - 2) + 2(1 + 6x)$??? $6x - 2$

(b) $2(-7 - x)(-2x - 3) - 5(3 + 2x)$??? $4x^2 - 24x + 27$

(c) $(x - 2)^3 - (3x + 3)^2$??? $-x^3 + 15x^2 - 17$

(d) $\frac{x + 4}{5} - 2 \cdot \frac{-1 + 5x}{2}$??? $\frac{18}{-10}$

(e) $\frac{\frac{-1}{-1} - \frac{3}{x}}{\frac{1}{8} + \frac{-1}{1}}$??? $\frac{-8x + 24}{7x}$

(f) $\frac{(1 - 4x)^2 + 6}{(8x - 9) \cdot \frac{7}{x}}$??? $\frac{16x^3 - 8x^2 - 7x}{-56x - 63}$

2.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Omicron o* -iii

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

(a) $5(5x - 3) + 1(4 + x)$??? $26x - 11$

(b) $-5(-1 + 4x)(3x + 1) + 2(-1 - 3x)$??? $-60x^2 + 11x$

(c) $(3x - 2)^3 - (-4x + 2)^2$??? $-27x^3 - 70x^2 + 52x - 12$

(d) $\frac{9x - 1}{-4} - 2 \cdot \frac{-2 + 2x}{-3}$??? $\frac{-11x - 13}{-12}$

(e) $\frac{\frac{-1}{-1} - \frac{-1}{x}}{\frac{1}{1} + \frac{8}{-3}}$??? $\frac{4x + 1}{-5x}$

(f) $\frac{(1 - 8x)^2 - 3}{(6x - 2) \cdot \frac{8}{x}}$??? $\frac{64x^3 - 16x^2 + 2x}{-48x - 16}$

3.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Omicron o* -iv

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

(a) $5(2x + 1) - 2(-1 + 2x)$??? $6x$

(b) $-4(1 + 8x)(3x - 1) - 1(-5 + 2x)$??? $-96x^2 + 18x + 9$

(c) $(-3x - 1)^3 - (-2x + 2)^2$. ??? . $-27x^3 - 31x^2 - x - 5$

(d) $\frac{-4x + 1}{3} - 5 \cdot \frac{-2 - x}{3}$??? $\frac{-3x + 33}{-9}$

(e) $\frac{\frac{-4}{-3} - \frac{-3}{x}}{\frac{1}{-1} + \frac{-4}{-7}}$??? $\frac{-28x - 63}{9x}$

(f) $\frac{(-2 - 6x)^2 + 4}{(-4x + 1) \cdot \frac{-5}{x}}$??? $\frac{36x^3 + 24x^2 + 8x}{20x - 5}$

4.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group $Pi \pi$ -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-5(-2x - 5) - 8(-1 - x) \dots ??? \dots 18x + 33$

(b) $2(1 + 5x)(3x + 9) + 9(-2 + 3x) ??? 30x^2 - 123x$

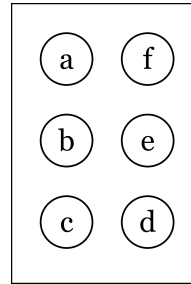
(c) $(2x - 3)^3 - (5x + 3)^2 \dots ??? \dots -8x^3 + 61x^2 - 36$

(d) $\frac{2x - 6}{-5} - 4 \cdot \frac{-2 + 4x}{-4} \dots ??? \dots \frac{-72x - 16}{-20}$

(e) $\frac{\frac{1}{3} - \frac{-1}{x}}{\frac{1}{3} + \frac{-7}{-1}} \dots ??? \dots \frac{-3x - 3}{-22x}$

(f) $\frac{(4 - 2x)^2 - 4}{(6x + 3) \cdot \frac{4}{x}} \dots ??? \dots \frac{4x^3 - 16x^2 + 12x}{24x + 12}$

1.



Braille letter

Simplifying an Expression, group $Pi \pi$ -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $4(-3x - 2) - 2(1 + 6x) \dots ??? \dots -24x - 10$

(b) $2(1 - 4x)(x - 2) - 6(3 - x) ??? -8x^2 - 24x - 22$

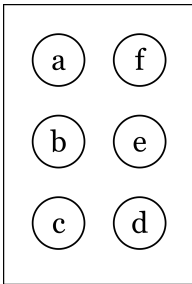
(c) $(-x + 1)^3 - (-6x - 1)^2 \dots ??? \dots x^3 + 33x^2 - 15x$

(d) $\frac{3x + 6}{6} - 3 \cdot \frac{3 + 3x}{8} \dots ??? \dots \frac{30x - 6}{-48}$

(e) $\frac{\frac{1}{-6} - \frac{2}{x}}{\frac{1}{3} + \frac{-3}{-1}} \dots ??? \dots \frac{-38}{60x}$

(f) $\frac{(1 + 3x)^2 + 4}{(x - 7) \cdot \frac{-3}{x}} \dots ??? \dots \frac{9x^3 + 6x^2 - 5x}{3x + 21}$

2.



Braille letter

Simplifying an Expression, group $Pi \pi$ -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $2(-7x - 2) + 1(-6 - 4x) \dots ??? \dots -18x$

(b) $-3(-2 - 3x)(-5x - 6) - 3(-8 + 4x) ??? -45x^2 - 96x - 12$

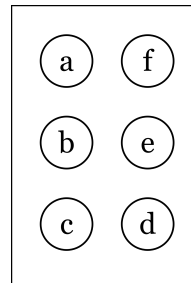
(c) $(-2x - 1)^3 - (-5x - 2)^2 \dots ??? \dots -8x^3 - 37x^2 - 26x - 5$

(d) $\frac{3x + 7}{-3} - 5 \cdot \frac{-2 - x}{5} \dots ??? \dots \frac{5}{15}$

(e) $\frac{\frac{3}{-4} - \frac{-1}{x}}{\frac{1}{1} + \frac{-1}{6}} \dots ??? \dots \frac{18x - 24}{-20x}$

(f) $\frac{(2 + 5x)^2 - 1}{(-9x + 2) \cdot \frac{7}{x}} \dots ??? \dots \frac{25x^3 + 20x^2 + 3x}{-63x + 14}$

3.



Braille letter

Simplifying an Expression, group $Pi \pi$ -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $8(-8x + 9) - 3(-5 - 4x) \dots ??? \dots -52x + 87$

(b) $-3(-7 + 6x)(-x + 1) - 6(-4 + x) ??? 18x^2 + 45x - 45$

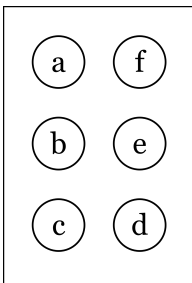
(c) $(-4x - 2)^3 - (-2x - 1)^2 \dots ??? \dots 64x^3 + 100x^2 - 52x - 9$

(d) $\frac{-x - 6}{4} + 7 \cdot \frac{-1 + x}{5} \dots ??? \dots \frac{-23x - 58}{-20}$

(e) $\frac{\frac{5}{-3} - \frac{-2}{x}}{\frac{1}{-7} + \frac{1}{2}} \dots ??? \dots \frac{-70x + 84}{15x}$

(f) $\frac{(6 - 7x)^2 + 2}{(-5x - 2) \cdot \frac{-3}{x}} \dots ??? \dots \frac{49x^3 - 84x^2 - 38x}{-15x + 6}$

4.



Braille letter

Simplifying an Expression, group *Rho* ρ -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $5(-9x + 2) + 5(1 - 2x)$??? $-55x + 15$

(b) $3(4 - 3x)(-3x + 6) - 3(9 + 2x)$.. ??? .. $27x^2 + 96x$

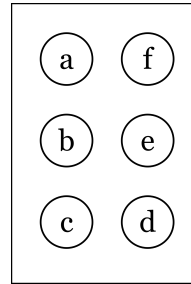
(c) $(-3x + 2)^3 - (-3x - 6)^2$??? $27x^3 - 45x^2 - 72x - 28$

(d) $\frac{4x - 3}{5} - 3 \cdot \frac{-3 + x}{6}$??? $\frac{9x + 27}{-30}$

(e) $\frac{\frac{-9}{-3} - \frac{1}{x}}{\frac{1}{4} + \frac{8}{2}}$??? $\frac{-72x + 24}{-102x}$

(f) $\frac{(1 - 6x)^2 - 1}{(-5x + 1) \cdot \frac{-1}{x}}$??? $\frac{36x^3 - 12x^2}{-5x - 1}$

1.



Braille letter

Simplifying an Expression, group *Rho* ρ -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-3(5x - 1) + 2(1 - 8x)$??? $-31x + 5$

(b) $5(-2 - 3x)(x + 4) - 3(8 - 4x)$??? $-15x^2 + 58x - 64$

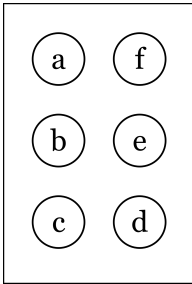
(c) $(x + 1)^3 - (-2x - 3)^2$... ??? ... $-x^3 - x^2 - 9x - 8$

(d) $\frac{x + 2}{3} + 5 \cdot \frac{1 - 5x}{6}$??? $\frac{-69x + 27}{-18}$

(e) $\frac{\frac{9}{-1} - \frac{-5}{x}}{\frac{1}{3} + \frac{-4}{3}}$??? $\frac{84x - 46}{9x}$

(f) $\frac{(-3 - 4x)^2 + 4}{(3x - 5) \cdot \frac{3}{x}}$??? $\frac{16x^3 + 24x^2 - 13x}{-9x - 15}$

2.



Braille letter

Simplifying an Expression, group *Rho* ρ -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $3(-5x + 7) + 3(-1 + 8x)$??? $9x$

(b) $3(-5 - 9x)(x - 2) + 6(5 + 4x)$??? $-27x^2 + 63x + 60$

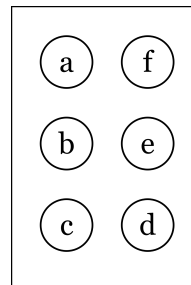
(c) $(2x + 3)^3 - (4x - 5)^2$.. ??? .. $8x^3 + 20x^2 + 94x + 2$

(d) $\frac{-8x - 6}{-4} + 3 \cdot \frac{3 - 2x}{9}$??? $\frac{-90}{36}$

(e) $\frac{\frac{3}{-3} - \frac{5}{x}}{\frac{1}{3} + \frac{5}{-2}}$??? $\frac{-19x - 89}{-39x}$

(f) $\frac{(-1 - 9x)^2 - 8}{(x + 1) \cdot \frac{1}{x}}$??? $\frac{81x^3 + 18x^2 - 7x}{x + 1}$

3.



Braille letter

Simplifying an Expression, group *Rho* ρ -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $9(3x - 1) - 3(2 - 2x)$??? $33x + 15$

(b) $-2(-1 - 3x)(4x + 4) + 8(-6 - x)$??? $24x^2 + 24x - 40$

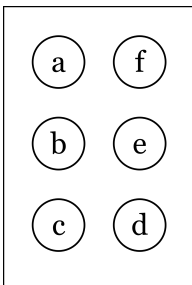
(c) $(-x + 2)^3 - (-9x + 1)^2$.. ??? .. $-x^3 - 75x^2 + 6x + 7$

(d) $\frac{7x + 6}{5} + 2 \cdot \frac{1 + 5x}{-7}$??? $\frac{-x - 32}{35}$

(e) $\frac{\frac{-3}{-1} - \frac{5}{x}}{\frac{1}{-1} + \frac{-4}{1}}$??? $\frac{3x - 5}{-5x}$

(f) $\frac{(-4 + 2x)^2 + 4}{(-9x - 3) \cdot \frac{-5}{x}}$??? $\frac{4x^3 - 16x^2 + 20x}{45x + 15}$

4.



Braille letter

Simplifying an Expression, group *Sigma* σ -i

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

- (a) $-8(9x + 4) - 8(6 - 4x)$??? $-40x - 80$
- (b) $4(-1 - 5x)(5x + 4) - 4(-2 - 7x)$??? $-100x^2 - 72x - 8$
- (c) $(2x + 4)^3 - (x - 4)^2$... ??? ... $8x^3 + 47x^2 + 104x + 48$
- (d) $\frac{-2x + 7}{4} + 2 \cdot \frac{7 - 2x}{5}$??? $\frac{26x + 91}{-20}$
- (e) $\frac{\frac{-3}{-3} - \frac{-1}{x}}{\frac{1}{-4} + \frac{1}{-3}}$??? $\frac{-36x - 36}{21x}$
- (f) $\frac{(-8 - 6x)^2 + 6}{(7x - 5) \cdot \frac{-9}{x}}$??? $\frac{36x^3 + 96x^2 - 70x}{63x + 45}$

1.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Sigma* σ -ii

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

- (a) $7(7x + 6) - 2(-4 - 2x)$??? $53x + 50$
- (b) $-3(3 + 3x)(3x - 1) - 1(3 + 8x)$??? $-27x^2 + 26x$
- (c) $(x + 4)^3 - (-2x + 2)^2$. ??? . $x^3 + 8x^2 + 56x + 60$
- (d) $\frac{4x - 4}{3} + 6 \cdot \frac{5 - x}{-8}$??? $\frac{-50x + 122}{24}$
- (e) $\frac{\frac{-1}{-2} - \frac{2}{x}}{\frac{1}{2} + \frac{1}{-1}}$??? $\frac{2x - 8}{-2x}$
- (f) $\frac{(-2 + 3x)^2 - 4}{(-2x + 3) \cdot \frac{2}{x}}$??? $\frac{9x^3 - 12x^2}{4x + 6}$

2.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Sigma* σ -iii

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

- (a) $5(2x + 3) + 6(-1 + x)$??? $16x + 9$
- (b) $2(6 + x)(-6x + 1) - 5(-3 - 2x)$??? $-12x^2 + 60x$
- (c) $(x - 3)^3 - (-x + 1)^2$. ??? . $x^3 - 10x^2 + 29x - 28$
- (d) $\frac{-2x + 3}{6} - 2 \cdot \frac{-8 + 4x}{-6}$??? $\frac{-36x + 78}{36}$
- (e) $\frac{\frac{4}{-3} - \frac{1}{x}}{\frac{1}{7} + \frac{-4}{-1}}$??? $\frac{-28x - 21}{87x}$
- (f) $\frac{(-2 - 2x)^2 - 1}{(-7x + 1) \cdot \frac{1}{x}}$??? $\frac{4x^3 + 8x^2 - 3x}{7x + 1}$

3.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Sigma* σ -iv

Name:

- (a) Expand the brackets and simplify the expression.
- (b) Expand the brackets and simplify the expression.
- (c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.
- (e) Simplify the expression.
- (f) Simplify the expression.

- (a) $-6(-2x + 4) + 6(-3 + 3x)$??? $30x - 42$
- (b) $-2(-3 + 4x)(-8x - 1) + 4(6 + 3x)$. ??? . $64x^2 + 28x$
- (c) $(-4x - 4)^3 - (x - 2)^2$??? $-64x^3 - 193x^2 - 188x - 68$
- (d) $\frac{-3x + 6}{-2} - 3 \cdot \frac{6 + x}{7}$??? $\frac{78}{14}$
- (e) $\frac{\frac{4}{5} - \frac{-3}{x}}{\frac{1}{-2} + \frac{-1}{2}}$??? $\frac{-17x - 59}{20x}$
- (f) $\frac{(3 - x)^2 + 2}{(-6x + 3) \cdot \frac{1}{x}}$??? $\frac{x^3 - 6x^2 + 11x}{-6x + 3}$

4.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Tau* τ -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-2(-9x - 8) - 7(-4 - 8x) \dots \dots \dots ??? \dots \dots 74x - 44$

(b) $-2(3 - x)(7x + 1) + 1(-5 + x) \dots \dots \dots ??? \dots \dots 14x^2 - 39x - 11$

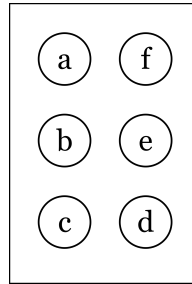
(c) $(4x - 1)^3 - (7x + 6)^2 \dots \dots \dots ??? \dots \dots 64x^3 - 97x^2 - 72x - 37$

(d) $\frac{-3x + 1}{5} + 2 \cdot \frac{4 - x}{-5} \dots \dots \dots ??? \dots \dots \frac{5x + 35}{25}$

(e) $\frac{\frac{-1}{3} - \frac{2}{x}}{\frac{1}{1} + \frac{-5}{-6}} \dots \dots \dots ??? \dots \dots \frac{6x + 36}{-33x}$

(f) $\frac{(-2 + x)^2 - 5}{(x + 8) \cdot \frac{1}{x}} \dots \dots \dots ??? \dots \dots \frac{x^3 - 4x^2 - x}{x + 8}$

1.



Braille letter

Simplifying an Expression, group *Tau* τ -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $7(2x + 1) - 6(-2 + 2x) \dots \dots \dots ??? \dots \dots 2x + 19$

(b) $-2(5 - x)(-6x - 1) - 4(2 - x) \dots \dots \dots ??? \dots \dots -12x^2 + 62x + 2$

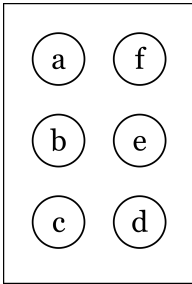
(c) $(x - 1)^3 - (2x + 3)^2 \dots \dots \dots ??? \dots \dots x^3 - 7x^2 - 9x - 10$

(d) $\frac{-2x - 8}{2} + 9 \cdot \frac{5 - 2x}{-5} \dots \dots \dots ??? \dots \dots \frac{-26x + 130}{10}$

(e) $\frac{\frac{-7}{3} - \frac{6}{x}}{\frac{1}{1} + \frac{-8}{-5}} \dots \dots \dots ??? \dots \dots \frac{35x + 90}{-39x}$

(f) $\frac{(2 - x)^2 - 1}{(4x + 2) \cdot \frac{-7}{x}} \dots \dots \dots ??? \dots \dots \frac{x^3 - 4x^2 - 3x}{28x - 14}$

2.



Braille letter

Simplifying an Expression, group *Tau* τ -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $5(-2x + 7) + 6(1 - 7x) \dots \dots \dots ??? \dots \dots -52x + 41$

(b) $3(6 - 2x)(2x + 2) + 1(-7 - 9x) \dots \dots \dots ??? \dots \dots -12x^2 - 15x$

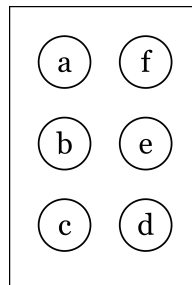
(c) $(-3x + 1)^3 - (-4x + 4)^2 \dots \dots \dots ??? \dots \dots 27x^3 - 11x^2 + 23x - 15$

(d) $\frac{-3x - 4}{-4} + 2 \cdot \frac{2 - 5x}{-2} \dots \dots \dots ??? \dots \dots \frac{-46x - 8}{-8}$

(e) $\frac{\frac{2}{-4} - \frac{-1}{x}}{\frac{1}{4} + \frac{-1}{-5}} \dots \dots \dots ??? \dots \dots \frac{-40x + 80}{36x}$

(f) $\frac{(1 + 8x)^2 - 5}{(-x + 1) \cdot \frac{3}{x}} \dots \dots \dots ??? \dots \dots \frac{64x^3 + 16x^2 + 4x}{3x + 3}$

3.



Braille letter

Simplifying an Expression, group *Tau* τ -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-8(-x - 2) - 4(3 - 8x) \dots \dots \dots ??? \dots \dots 40x + 4$

(b) $2(-6 + 2x)(-8x + 3) - 3(-3 + 5x) \dots \dots \dots ??? \dots \dots -32x^2 - 93x + 27$

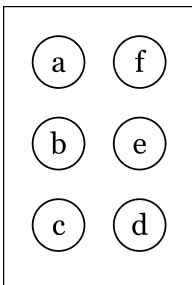
(c) $(3x + 1)^3 - (7x - 1)^2 \dots \dots \dots ??? \dots \dots -27x^3 + 22x^2 + 23x$

(d) $\frac{2x - 4}{6} + 2 \cdot \frac{4 - x}{5} \dots \dots \dots ??? \dots \dots \frac{28}{-30}$

(e) $\frac{\frac{5}{-2} - \frac{1}{x}}{\frac{1}{-1} + \frac{-1}{-4}} \dots \dots \dots ??? \dots \dots \frac{20x + 8}{6x}$

(f) $\frac{(5 - 2x)^2 + 3}{(x + 8) \cdot \frac{4}{x}} \dots \dots \dots ??? \dots \dots \frac{4x^3 - 20x^2 - 28x}{-4x + 32}$

4.



Braille letter

Simplifying an Expression, group *Upsilon* v -i

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

(a) $-3(5x + 1) + 3(-4 + 2x)$??? $-9x - 15$

(b) $3(-1 + 3x)(8x - 4) + 2(-5 + 4x)$??? $72x^2 - 52x + 2$

(c) $(-2x - 4)^3 - (-2x + 5)^2$... ??? ... $8x^3 - 52x^2 - 89$

(d) $\frac{-3x - 6}{-3} + 7 \cdot \frac{-1 + 4x}{6}$??? $\frac{-102x - 15}{18}$

(e) $\frac{\frac{2}{-3} - \frac{-2}{x}}{\frac{1}{4} + \frac{-4}{-3}}$??? $\frac{-22x + 69}{57x}$

(f) $\frac{(-5 - 4x)^2 - 7}{(6x - 8) \cdot \frac{1}{x}}$??? $\frac{16x^3 + 40x^2 - 18x}{-6x - 8}$

1.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Upsilon* v -ii

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

(a) $-8(3x + 3) - 5(9 - 4x)$??? $-4x - 69$

(b) $-2(4 + 4x)(-x - 4) - 5(3 + 4x)$... ??? ... $8x^2 - 20x$

(c) $(-2x + 4)^3 - (-3x + 5)^2$??? $-8x^3 + 39x^2 - 66x + 39$

(d) $\frac{-2x - 3}{-3} - 2 \cdot \frac{-4 + 4x}{-3}$??? $\frac{-30x - 15}{-9}$

(e) $\frac{\frac{7}{-3} - \frac{-1}{x}}{\frac{1}{1} + \frac{-3}{-8}}$??? $\frac{-56x + 24}{33x}$

(f) $\frac{(-2 - 2x)^2 + 2}{(4x - 1) \cdot \frac{5}{x}}$??? $\frac{4x^3 + 8x^2 - 6x}{-20x - 5}$

2.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Upsilon* v -iii

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

(a) $2(-x - 3) + 1(-1 - 4x)$??? $-6x - 7$

(b) $2(4 - 3x)(-8x + 3) + 2(4 - x)$... ??? ... $48x^2 + 84x$

(c) $(3x - 3)^3 - (-4x - 3)^2$??? $-27x^3 + 97x^2 + 57x - 36$

(d) $\frac{7x - 5}{-5} + 3 \cdot \frac{1 - 3x}{-4}$??? $\frac{5}{-20}$

(e) $\frac{\frac{-1}{-2} - \frac{-2}{x}}{\frac{1}{1} + \frac{-2}{4}}$??? $\frac{-4x - 16}{-4x}$

(f) $\frac{(9 - x)^2 - 4}{(2x - 1) \cdot \frac{4}{x}}$??? $\frac{x^3 - 18x^2 + 77x}{8x - 4}$

3.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Upsilon* v -iv

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

(a) $-8(x + 2) - 2(-6 - 5x)$??? $2x - 4$

(b) $-2(-4 + 2x)(x - 1) + 5(-2 + 9x)$??? $-4x^2 - 57x - 18$

(c) $(4x - 1)^3 - (5x - 3)^2$.. ??? .. $64x^3 - 73x^2 + 42x - 10$

(d) $\frac{-5x - 1}{3} + 8 \cdot \frac{-4 - 3x}{-5}$??? $\frac{-47x - 91}{-15}$

(e) $\frac{\frac{-4}{-5} - \frac{5}{x}}{\frac{1}{-1} + \frac{6}{-3}}$??? $\frac{-12x + 75}{45x}$

(f) $\frac{(-4 + x)^2 - 5}{(-2x + 1) \cdot \frac{4}{x}}$??? $\frac{x^3 - 8x^2 + 11x}{-8x + 4}$

4.

a

f

b

e

c

d

Braille letter

Simplifying an Expression, group *Phi* ϕ -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-6(-3x - 1) - 8(5 + 3x)$??? $-6x - 34$

(b) $-2(-5 + 2x)(-3x - 1) + 3(-8 - x)$??? $12x^2 - 29x - 34$

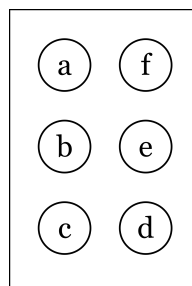
(c) $(-2x + 2)^3 - (7x + 6)^2$??? $8x^3 - 25x^2 - 28$

(d) $\frac{5x - 6}{3} - 4 \cdot \frac{-1 + 3x}{-7}$??? $\frac{71x + 54}{21}$

(e) $\frac{\frac{-5}{-4} - \frac{-4}{x}}{\frac{1}{-4} + \frac{-2}{-1}}$??? $\frac{-17x - 67}{-28x}$

(f) $\frac{(1 - 6x)^2 - 7}{(-8x - 2) \cdot \frac{6}{x}}$??? $\frac{36x^3 - 12x^2 - 6x}{-48x - 12}$

1.



Braille letter

Simplifying an Expression, group *Phi* ϕ -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-9(4x - 4) - 1(-1 + 3x)$??? $-39x + 37$

(b) $3(-3 - 6x)(-4x + 5) + 2(1 - 9x)$. ??? . $72x^2 + 72x$

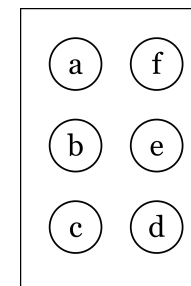
(c) $(4x + 4)^3 - (x + 1)^2$??? $-64x^3 + 191x^2 + 190x + 63$

(d) $\frac{3x + 6}{8} + 2 \cdot \frac{1 + 6x}{-9}$??? $\frac{-69x - 38}{72}$

(e) $\frac{\frac{4}{1} - \frac{1}{x}}{\frac{1}{-4} + \frac{4}{-5}}$??? $\frac{80x - 20}{-21x}$

(f) $\frac{(4 + 2x)^2 + 2}{(x + 2) \cdot \frac{4}{x}}$??? $\frac{4x^3 + 16x^2 - 18x}{-4x + 8}$

2.



Braille letter

Simplifying an Expression, group *Phi* ϕ -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $7(-6x - 5) + 2(-4 + 8x)$??? $-26x - 43$

(b) $-2(-6 + x)(6x + 4) - 7(-3 - 3x)$??? $-12x^2 - 85x + 69$

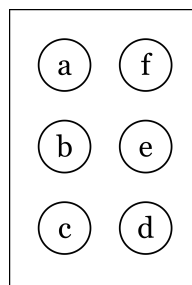
(c) $(-3x - 3)^3 - (4x + 5)^2$.. ??? .. $27x^3 + 97x^2 - 121x - 52$

(d) $\frac{-2x + 3}{-3} + 5 \cdot \frac{-2 - 4x}{6}$??? $\frac{48x + 48}{18}$

(e) $\frac{\frac{3}{-3} - \frac{5}{x}}{\frac{1}{1} + \frac{-2}{4}}$??? $\frac{12x + 60}{-6x}$

(f) $\frac{(-2 - 3x)^2 - 6}{(4x + 1) \cdot \frac{7}{x}}$??? $\frac{9x^3 + 12x^2 + 2x}{-28x + 7}$

3.



Braille letter

Simplifying an Expression, group *Phi* ϕ -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-2(-3x + 4) - 7(4 - x)$??? $13x + 36$

(b) $4(-6 + 8x)(-x + 2) + 1(-4 - 3x)$??? $-32x^2 + 85x - 52$

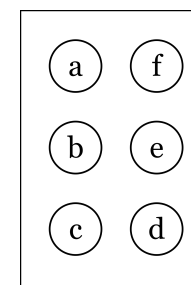
(c) $(4x - 4)^3 - (2x - 8)^2$. ??? . $64x^3 - 196x^2 + 224x - 128$

(d) $\frac{-3x + 2}{6} + 2 \cdot \frac{7 - 4x}{-7}$??? $\frac{-27x + 70}{42}$

(e) $\frac{\frac{-1}{-2} - \frac{1}{x}}{\frac{1}{-1} + \frac{-8}{-4}}$??? $\frac{-4x + 8}{-8x}$

(f) $\frac{(-3 + 2x)^2 - 2}{(2x + 2) \cdot \frac{-5}{x}}$??? $\frac{4x^3 - 12x^2 + 7x}{-10x - 10}$

4.



Braille letter

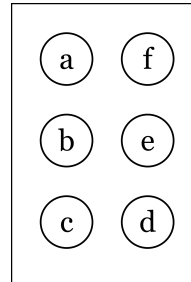
Simplifying an Expression, group *Chi* χ -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $6(x-1) + 3(-2+x)$??? $9x-12$
 (b) $-5(6+6x)(-2x-3) - 4(-2+4x)$??? $60x^2 + 134x + 98$
 (c) $(3x-4)^3 - (-2x-7)^2$. ??? . $27x^3 - 112x^2 + 116x - 113$
 (d) $\frac{-3x+1}{3} + 4 \cdot \frac{1-3x}{4}$??? $\frac{-48x+16}{12}$
 (e) $\frac{\frac{2}{1} - \frac{-3}{x}}{\frac{1}{6} + \frac{1}{-2}}$??? $\frac{-24x-36}{4x}$
 (f) $\frac{(-5-4x)^2 + 6}{(-2x-6) \cdot \frac{1}{x}}$??? $\frac{16x^3 + 40x^2 - 31x}{2x-6}$

1.



Braille letter

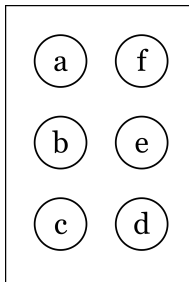
Simplifying an Expression, group *Chi* χ -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $3(4x-8) - 4(3-3x)$??? $24x-36$
 (b) $2(-3+4x)(3x+7) - 1(-5+5x)$??? $24x^2 - 33x - 37$
 (c) $(3x-4)^3 - (x+4)^2$??? $-27x^3 + 109x^2 + 136x - 80$
 (d) $\frac{-3x-8}{5} + 3 \cdot \frac{2-5x}{9}$??? $\frac{-42}{-45}$
 (e) $\frac{\frac{9}{-1} - \frac{-1}{x}}{\frac{1}{2} + \frac{6}{-2}}$??? $\frac{-35x+3}{-10x}$
 (f) $\frac{(2-x)^2 - 7}{(-x+4) \cdot \frac{6}{x}}$??? $\frac{x^3 - 4x^2 + 3x}{6x+24}$

2.



Braille letter

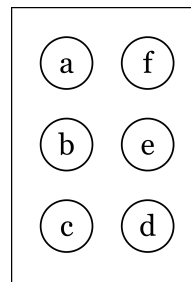
Simplifying an Expression, group *Chi* χ -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $-4(4x+3) - 3(-7-6x)$??? $2x+9$
 (b) $3(2+4x)(8x-6) - 3(4+x)$.. ??? .. $96x^2 - 27x - 48$
 (c) $(-3x-2)^3 - (-7x-1)^2$??? $-27x^3 - 103x^2 - 50x - 9$
 (d) $\frac{-3x+1}{4} + 7 \cdot \frac{-2-2x}{-5}$??? $\frac{-41x-61}{20}$
 (e) $\frac{\frac{-2}{-1} - \frac{-6}{x}}{\frac{1}{-1} + \frac{-2}{-3}}$??? $\frac{-8x-19}{x}$
 (f) $\frac{(3-x)^2 + 3}{(-x-5) \cdot \frac{2}{x}}$??? $\frac{x^3 - 6x^2 - 12x}{2x-10}$

3.



Braille letter

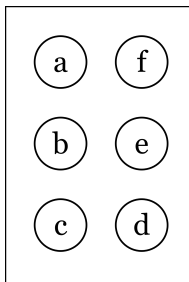
Simplifying an Expression, group *Chi* χ -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

- (a) $-5(-9x-5) - 5(7+3x)$??? $30x-10$
 (b) $-4(3+9x)(-x+2) + 9(-1-5x)$. ??? . $36x^2 + 105x$
 (c) $(-3x+1)^3 - (3x+4)^2$??? $-27x^3 + 18x^2 - 33x - 15$
 (d) $\frac{-6x-5}{3} - 7 \cdot \frac{1+3x}{4}$??? $\frac{-87x-41}{-12}$
 (e) $\frac{\frac{1}{1} - \frac{1}{x}}{\frac{1}{-2} + \frac{-8}{6}}$??? $\frac{-13x+11}{22x}$
 (f) $\frac{(3+4x)^2 + 2}{(8x+6) \cdot \frac{-5}{x}}$??? $\frac{16x^3 + 24x^2 - 11x}{40x-30}$

4.



Braille letter

Simplifying an Expression, group *Psi* ψ -i

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $2(-7x + 1) + 6(-1 - 4x)$??? $-38x - 4$

(b) $-2(1 - 8x)(-2x - 6) - 2(-4 - 2x)$??? $-32x^2 - 88x + 20$

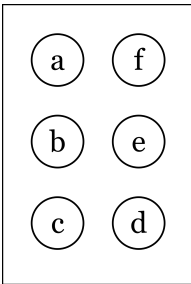
(c) $(2x + 1)^3 - (-5x + 7)^2$??? $-8x^3 - 13x^2 - 48$

(d) $\frac{-x - 9}{-2} - 2 \cdot \frac{7 + 4x}{-6}$??? $\frac{22x + 82}{-12}$

(e) $\frac{\frac{4}{-7} - \frac{-3}{x}}{\frac{1}{-1} + \frac{-2}{-1}}$??? $\frac{7x - 19}{-7x}$

(f) $\frac{(2 - 5x)^2 - 2}{(3x + 3) \cdot \frac{1}{x}}$??? $\frac{25x^3 - 20x^2 - 2x}{-3x + 3}$

1.



Braille letter

Simplifying an Expression, group *Psi* ψ -ii

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $3(2x + 7) + 1(2 + 4x)$??? $10x + 23$

(b) $4(4 - x)(-5x - 3) - 1(1 + x)$??? $20x^2 + 69x$

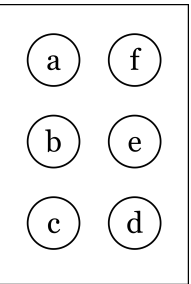
(c) $(-4x + 3)^3 - (-x - 9)^2$??? $64x^3 + 143x^2 - 54$

(d) $\frac{8x + 1}{2} + 4 \cdot \frac{1 + 3x}{8}$??? $\frac{16}{-16}$

(e) $\frac{\frac{-5}{-5} - \frac{-1}{x}}{\frac{1}{2} + \frac{1}{-6}}$??? $\frac{59x + 57}{20x}$

(f) $\frac{(-4 - 4x)^2 - 5}{(-2x + 6) \cdot \frac{-3}{x}}$... ??? ... $\frac{16x^3 + 32x^2 - 11x}{-6x - 18}$

2.



Braille letter

Simplifying an Expression, group *Psi* ψ -iii

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $5(-4x - 3) + 7(-3 + x)$??? $-13x - 36$

(b) $6(4 - 3x)(-2x + 1) - 3(6 - 7x)$??? $36x^2 - 45x + 6$

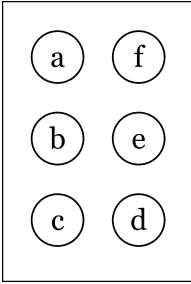
(c) $(-4x - 3)^3 - (-7x - 3)^2$. ??? . $64x^3 - 193x^2 - 36$

(d) $\frac{-8x - 3}{3} - 6 \cdot \frac{3 - 5x}{-8}$??? $\frac{-30}{24}$

(e) $\frac{\frac{-1}{-1} - \frac{4}{x}}{\frac{1}{-2} + \frac{7}{-8}}$??? $\frac{-18x + 67}{22x}$

(f) $\frac{(-3 + 3x)^2 + 7}{(-8x + 1) \cdot \frac{-3}{x}}$??? $\frac{9x^3 - 18x^2 - 16x}{-24x - 3}$

3.



Braille letter

Simplifying an Expression, group *Psi* ψ -iv

Name:

- (a) Expand the brackets and simplify the expression.

(b) Expand the brackets and simplify the expression.

(c) Expand the brackets and simplify the expression.
- (d) Simplify the expression.

(e) Simplify the expression.

(f) Simplify the expression.

- (a) $3(x + 6) + 7(-4 + x)$??? $10x - 10$

(b) $-2(-1 + 6x)(-4x - 4) + 6(-9 + 4x)$??? $48x^2 - 64x$

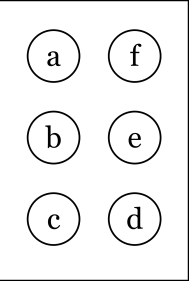
(c) $(4x - 2)^3 - (-2x + 6)^2$??? $64x^3 - 100x^2 + 72x - 44$

(d) $\frac{-7x - 8}{-8} + 3 \cdot \frac{4 - 2x}{-4}$??? $\frac{76x - 64}{32}$

(e) $\frac{\frac{-7}{1} - \frac{-8}{x}}{\frac{1}{-3} + \frac{2}{2}}$??? $\frac{42x - 48}{-4x}$

(f) $\frac{(3 - 6x)^2 - 5}{(3x - 3) \cdot \frac{1}{x}}$??? $\frac{36x^3 - 36x^2 + 4x}{3x - 3}$

4.



Braille letter

Simplifying an Expression, group *Omega* ω -i

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $8(2x - 1) + 4(-6 - 5x)$??? $-4x - 32$

(b) $3(-3 - 3x)(3x - 2) - 3(1 + 8x)$.. ??? .. $-27x^2 + 33x$

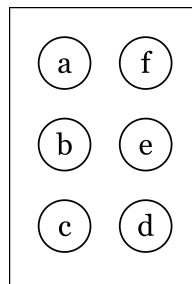
(c) $(-4x + 1)^3 - (8x + 5)^2$??? $-64x^3 - 16x^2 - 92x - 24$

(d) $\frac{4x - 1}{-3} + 2 \cdot \frac{6 - 4x}{4}$??? $\frac{40x - 40}{-12}$

(e) $\frac{\frac{1}{-1} - \frac{1}{x}}{\frac{1}{-2} + \frac{4}{-2}}$??? $\frac{5x + 5}{10x}$

(f) $\frac{(-9 - 4x)^2 - 1}{(8x + 5) \cdot \frac{4}{x}}$??? $\frac{16x^3 + 72x^2 - 80x}{-32x + 20}$

1.



Braille letter

Simplifying an Expression, group *Omega* ω -ii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $2(x - 3) - 8(-1 + 5x)$??? $-38x + 2$

(b) $-2(-3 + x)(-4x - 4) + 3(-2 - 2x)$??? $8x^2 + 22x - 30$

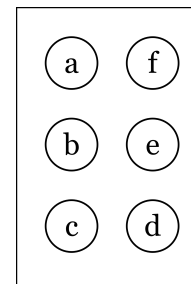
(c) $(-x - 1)^3 - (-x - 7)^2$... ??? ... $-x^3 - 4x^2 - 17x - 50$

(d) $\frac{3x - 1}{-4} - 4 \cdot \frac{-5 - 3x}{3}$??? $\frac{-83}{12}$

(e) $\frac{\frac{-1}{-7} - \frac{3}{x}}{\frac{1}{-1} + \frac{9}{-1}}$??? $\frac{-x + 21}{70x}$

(f) $\frac{(6 + x)^2 + 4}{(7x - 7) \cdot \frac{-3}{x}}$??? $\frac{x^3 + 12x^2 + 40x}{-21x + 21}$

2.



Braille letter

Simplifying an Expression, group *Omega* ω -iii

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-3(-7x - 4) + 2(1 - x)$??? $19x$

(b) $2(1 - 3x)(4x - 1) - 2(3 + 9x)$??? $-24x^2 - 4x - 8$

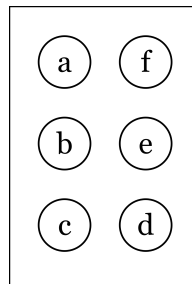
(c) $(x + 1)^3 - (-6x + 3)^2$??? $-x^3 - 33x^2 + 39x - 8$

(d) $\frac{6x + 3}{6} + 4 \cdot \frac{-1 + x}{-4}$??? $\frac{-36}{24}$

(e) $\frac{\frac{7}{-1} - \frac{4}{x}}{\frac{1}{1} + \frac{8}{2}}$??? $\frac{13x + 10}{-10x}$

(f) $\frac{(2 + 2x)^2 - 3}{(-8x - 6) \cdot \frac{-3}{x}}$??? $\frac{4x^3 + 8x^2 + x}{24x + 18}$

3.



Braille letter

Simplifying an Expression, group *Omega* ω -iv

Name:

- (a) Expand the brackets and simplify the expression. (d) Simplify the expression.
 (b) Expand the brackets and simplify the expression. (e) Simplify the expression.
 (c) Expand the brackets and simplify the expression. (f) Simplify the expression.

(a) $-7(4x + 5) - 2(-3 - 4x)$??? $-20x + 29$

(b) $2(-6 - 3x)(x - 6) + 3(-5 + 2x)$??? $-6x^2 + 30x + 57$

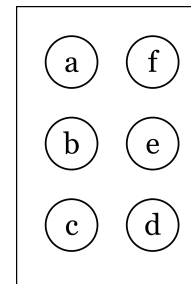
(c) $(x - 3)^3 - (6x + 3)^2$??? $x^3 - 45x^2 - 9x - 36$

(d) $\frac{x + 6}{7} + 4 \cdot \frac{-2 - 3x}{4}$??? $\frac{80x - 32}{-28}$

(e) $\frac{\frac{-6}{3} - \frac{-3}{x}}{\frac{1}{-7} + \frac{1}{1}}$??? $\frac{42x - 63}{-18x}$

(f) $\frac{(7 + 3x)^2 + 1}{(-x + 1) \cdot \frac{-5}{x}}$??? $\frac{9x^3 + 42x^2 + 50x}{5x - 5}$

4.



Braille letter

Simplifying an Expression (solutions)

α	$i : \mathbf{F}$	(a) $48x + 12\checkmark$	(b) $-60x^2 + 31x - 8\checkmark$	(c) $8x^3 - 57x^2 + 78x - 73\mathbf{X}$	(d) $\frac{3x+5}{-10}\mathbf{X}$	(e) $\frac{-4x-4}{2x}\mathbf{X}$	(f) $\frac{x^3-8x^2+11x}{30x-12}\checkmark$
	$ii : \mathbf{A}$	(a) $31x - 13\checkmark$	(b) $-48x^2 - 120x - 87\mathbf{X}$	(c) $-64x^3 - 112x^2 - 24x - 17\mathbf{X}$	(d) $\frac{37x+94}{-20}\mathbf{X}$	(e) $\frac{-12x+96}{28x}\mathbf{X}$	(f) $\frac{4x^3-20x^2+24x}{-12x+4}\mathbf{X}$
	$iii : \mathbf{C}$	(a) $-9x + 6\checkmark$	(b) $-20x^2 + 36x + 26\mathbf{X}$	(c) $x^3 - 15x^2 - 12\mathbf{X}$	(d) $\frac{-81x}{36}\mathbf{X}$	(e) $\frac{10x+10}{-30x}\mathbf{X}$	(f) $\frac{4x^3+24x^2+37x}{-8x+8}\checkmark$
	$iv : \mathbf{E}$	(a) $-20x - 13\checkmark$	(b) $42x^2 + 20x - 28\mathbf{X}$	(c) $x^3 - 22x^2 + 52x - 33\mathbf{X}$	(d) $\frac{-80x-32}{-32}\mathbf{X}$	(e) $\frac{36x+90}{-3x}\checkmark$	(f) $\frac{16x^3-64x^2+55x}{12x-21}\mathbf{X}$
β	$i : \mathbf{N}$	(a) $42x - 40\checkmark$	(b) $-48x^2 + 8x - 8\mathbf{X}$	(c) $x^3 - 33x^2 + 51x - 15\checkmark$	(d) $\frac{-39x-9}{18}\mathbf{X}$	(e) $\frac{18x+24}{30x}\checkmark$	(f) $\frac{36x^3+72x^2+27x}{-63x+72}\checkmark$
	$ii : \mathbf{A}$	(a) $-17x - 30\checkmark$	(b) $96x^2 - 108x + 22\mathbf{X}$	(c) $-27x^3 + 99x^2 - 126x + 55\mathbf{X}$	(d) $\frac{30x+18}{6}\mathbf{X}$	(e) $\frac{8x-48}{-54x}\mathbf{X}$	(f) $\frac{25x^3-50x^2+26x}{-24x+40}\mathbf{X}$
	$iii : \mathbf{M}$	(a) $-20x + 17\checkmark$	(b) $10x^2 - 30x - 21\mathbf{X}$	(c) $-8x^3 - 40x^2 - 62x - 31\checkmark$	(d) $\frac{48x-4}{8}\mathbf{X}$	(e) $\frac{-3x-18}{12x}\mathbf{X}$	(f) $\frac{4x^3-4x^2+5x}{8x-16}\checkmark$
	$iv : \mathbf{E}$	(a) $18x + 11\checkmark$	(b) $20x^2 - 34x + 14\mathbf{X}$	(c) $x^3 - 24x^2 + 36x + 63\mathbf{X}$	(d) $\frac{-83x+74}{-10}\mathbf{X}$	(e) $\frac{5x-70}{-32x}\checkmark$	(f) $\frac{25x^3-70x^2+54x}{-4x+8}\mathbf{X}$
γ	$i : \mathbf{D}$	(a) $-41x - 17\checkmark$	(b) $4x^2 + 16x + 20\mathbf{X}$	(c) $-x^3 - 13x^2 - 58x - 89\mathbf{X}$	(d) $\frac{-25x+6}{28}\mathbf{X}$	(e) $\frac{-10x-15}{-39x}\checkmark$	(f) $\frac{4x^3+4x^2+6x}{x-4}\checkmark$
	$ii : \mathbf{O}$	(a) $9x + 5\checkmark$	(b) $12x^2 - 50x + 22\mathbf{X}$	(c) $-64x^3 - 193x^2 - 198x - 73\checkmark$	(d) $\frac{4x-18}{4}\mathbf{X}$	(e) $\frac{14x+6}{-27x}\checkmark$	(f) $\frac{x^3+18x^2+82x}{24x-36}\mathbf{X}$
	$iii : \mathbf{O}$	(a) $-x - 6\checkmark$	(b) $32x^2 + 36\mathbf{X}$	(c) $-8x^3 + 11x^2 - 24x - 80\checkmark$	(d) $\frac{-47x-19}{-14}\mathbf{X}$	(e) $\frac{15x-81}{-9x}\checkmark$	(f) $\frac{9x^3+12x^2+5x}{-15x+12}\mathbf{X}$
	$iv : \mathbf{R}$	(a) $-13x - 37\checkmark$	(b) $-20x^2 + 12x + 8\checkmark$	(c) $8x^3 - 45x^2 + 18x - 63\checkmark$	(d) $\frac{48x-27}{-6}\mathbf{X}$	(e) $\frac{-4x+72}{-42x}\checkmark$	(f) $\frac{x^3-8x^2+18x}{x-2}\mathbf{X}$
δ	$i : \mathbf{A}$	(a) $-24\checkmark$	(b) $-8x^2 + 62x - 24\mathbf{X}$	(c) $8x^3 - 48x^2 - 54x - 26\mathbf{X}$	(d) $\frac{-14x-56}{-8}\mathbf{X}$	(e) $\frac{5x-72}{6x}\mathbf{X}$	(f) $\frac{x^3+6x^2+7x}{21x+42}\mathbf{X}$
	$ii : \mathbf{T}$	(a) $-6x + 1\mathbf{X}$	(b) $42x^2 + 28x - 21\checkmark$	(c) $-27x^3 - 28x^2 - 5x - 5\checkmark$	(d) $\frac{-47x-31}{21}\mathbf{X}$	(e) $\frac{-32x-80}{4x}\checkmark$	(f) $\frac{x^3+6x^2+15x}{24x-3}$
	$iii : \mathbf{O}$	(a) $-x - 6\checkmark$	(b) $32x^2 + 36\mathbf{X}$	(c) $-64x^3 - 73x^2 - 32x - 5\checkmark$	(d) $\frac{-44x-4}{-8}\mathbf{X}$	(e) $\frac{-48x-16}{-9x}\checkmark$	(f) $\frac{9x^3+12x^2+5x}{-15x+12}\mathbf{X}$
	$iv : \mathbf{M}$	(a) $29x - 36\checkmark$	(b) $-36x^2 + 9x + 25\mathbf{X}$	(c) $64x^3 + 183x^2 + 204x + 60\checkmark$	(d) $\frac{-51x-48}{15}\mathbf{X}$	(e) $\frac{3x-2}{30x-2}\mathbf{X}$	(f) $\frac{36x^3-72x^2+32x}{-15x+12}\checkmark$
ϵ	$i : \mathbf{S}$	(a) $-8x + 12\mathbf{X}$	(b) $80x^2 + 24x - 56\checkmark$	(c) $64x^3 - 49x^2 + 4x - 17\checkmark$	(d) $\frac{66x-30}{8}\mathbf{X}$	(e) $\frac{-27x-30}{40x}\mathbf{X}$	(f) $\frac{25x^3+10x^2-8x}{-5x+35}\checkmark$
	$ii : \mathbf{O}$	(a) $14x + 51\checkmark$	(b) $40x^2 - 66x + 28\mathbf{X}$	(c) $x^3 - 67x^2 + 35x - 5\checkmark$	(d) $\frac{-84x-70}{30}\mathbf{X}$	(e) $\frac{7x+6}{2x}\checkmark$	(f) $\frac{x^3+6x^2+5x}{x-4}\mathbf{X}$
	$iii : \mathbf{U}$	(a) $48x + 8\checkmark$	(b) $-28x^2 - 58x - 58\mathbf{X}$	(c) $-x^3 - 18x^2 - 9x - 36\checkmark$	(d) $\frac{87x-33}{-18}\checkmark$	(e) $\frac{16x+80}{40x}\mathbf{X}$	(f) $\frac{25x^3+70x^2+54x}{5x+5}\mathbf{X}$
	$iv : \mathbf{L}$	(a) $-18x + 27\checkmark$	(b) $-35x^2 - 12x + 23\checkmark$	(c) $x^3 - 16x^2 + 32x - 80\checkmark$	(d) $\frac{-44x-9}{10}\mathbf{X}$	(e) $\frac{72x+72}{-24x}\mathbf{X}$	(f) $\frac{64x^3+128x^2+70x}{-12x+48}\mathbf{X}$
ζ	$i : \mathbf{F}$	(a) $-42x + 14\checkmark$	(b) $8x^2 - 29x + 26\checkmark$	(c) $8x^3 + 39x^2 + 150x - 17\mathbf{X}$	(d) $\frac{-24x+20}{16}\mathbf{X}$	(e) $\frac{-6x-36}{-48x}\mathbf{X}$	(f) $\frac{64x^3+96x^2+32x}{12x+84}\checkmark$
	$ii : \mathbf{I}$	(a) $5x - 15\mathbf{X}$	(b) $-6x^2 - 24x - 6\checkmark$	(c) $-8x^3 + 32x^2 - 70x + 11\mathbf{X}$	(d) $\frac{-28x-20}{12}\mathbf{X}$	(e) $\frac{7x+24}{-42x}\mathbf{X}$	(f) $\frac{64x^3-32x^2+2x}{72x-48}\mathbf{X}$
	$iii : \mathbf{V}$	(a) $18x + 41\checkmark$	(b) $-6x^2 + 9x - 27\checkmark$	(c) $8x^3 + 8x^2 - 6x - 8\checkmark$	(d) $\frac{-64x-34}{-12}\checkmark$	(e) $\frac{32x+8}{-17x}\mathbf{X}$	(f) $\frac{9x^3-6x^2-2x}{9x-18}\mathbf{X}$
	$iv : \mathbf{E}$	(a) $27x + 4\checkmark$	(b) $-36x^2 - 68x - 80\mathbf{X}$	(c) $-27x^3 - 76x^2 - 79x - 26\mathbf{X}$	(d) $\frac{18x+42}{18}\mathbf{X}$	(e) $\frac{12x-48}{25x}\checkmark$	(f) $\frac{9x^3-42x^2+50x}{-21x-49}\mathbf{X}$
η	$i : \mathbf{G}$	(a) $21x - 9\checkmark$	(b) $80x^2 - 43x - 37\checkmark$	(c) $-27x^3 + 104x^2 - 132x + 55\mathbf{X}$	(d) $\frac{-x-23}{18}\mathbf{X}$	(e) $\frac{2x+7}{7x}\checkmark$	(f) $\frac{x^3+6x^2+8x}{12x+8}\checkmark$
	$ii : \mathbf{A}$	(a) $-5x + 1\checkmark$	(b) $30x^2 + 45x - 21\mathbf{X}$	(c) $-8x^3 - 28x^2 - 38x - 17\mathbf{X}$	(d) $\frac{36x+24}{-36}\mathbf{X}$	(e) $\frac{-16x-16}{9x}\mathbf{X}$	(f) $\frac{25x^3+30x^2+15x}{-15x-12}\mathbf{X}$
	$iii : \mathbf{M}$	(a) $-4x - 32\checkmark$	(b) $-60x^2 + 87x - 45\mathbf{X}$	(c) $x^3 - x^2 - x\checkmark$	(d) $\frac{12x+74}{-8}\mathbf{X}$	(e) $\frac{7x+36}{24x}\mathbf{X}$	(f) $\frac{x^3-19x^2+30x}{-2x+6}\checkmark$
	$iv : \mathbf{E}$	(a) $-5x + 25\checkmark$	(b) $2x^2 - 5x + 21\mathbf{X}$	(c) $-64x^3 + 60x^2 - 96x - 8\mathbf{X}$	(d) $\frac{48}{-48}\mathbf{X}$	(e) $\frac{18x+54}{-24x}\checkmark$	(f) $\frac{4x^3+20x^2+31x}{-10x+35}\mathbf{X}$
θ	$i : \mathbf{G}$	(a) $-4\checkmark$	(b) $90x^2 + 44x - 26\checkmark$	(c) $-x^3 - 72x^2 - 135x - 9\mathbf{X}$	(d) $\frac{54x+48}{-9}\mathbf{X}$	(e) $\frac{-4x+10}{-5x}\checkmark$	(f) $\frac{36x^3-48x^2+11x}{-10x+6}\checkmark$
	$ii : \mathbf{O}$	(a) $-39x + 24\checkmark$	(b) $-16x^2 - 12x + 16\mathbf{X}$	(c) $-8x^3 - 100x^2 - 182x - 91\checkmark$	(d) $\frac{-100x-12}{32}\mathbf{X}$	(e) $\frac{12x-84}{-26x}\checkmark$	(f) $\frac{x^3-24x^2+5x}{-24x-15}\mathbf{X}$
	$iii : \mathbf{L}$	(a) $-5x - 28\checkmark$	(b) $-48x^2 + 24x + 60\checkmark$	(c) $-64x^3 - 100x^2 - 64x - 24\checkmark$	(d) $\frac{-33x+9}{-6}\mathbf{X}$	(e) $\frac{-6x+2}{7x}\mathbf{X}$	(f) $\frac{9x^3-6x^2+3x}{9x+63}\mathbf{X}$
	$iv : \mathbf{D}$	(a) $6x + 32\checkmark$	(b) $84x^2 - 29x - 43\mathbf{X}$	(c) $27x^3 - 90x^2 + 99x - 36\mathbf{X}$	(d) $\frac{-32x+16}{42}\mathbf{X}$	(e) $\frac{-10x-75}{18x}\checkmark$	(f) $\frac{x^3+10x^2+27x}{2x-3}\checkmark$
ι	$i : \mathbf{L}$	(a) $18x - 8\checkmark$	(b) $60x^2 - 107x + 49\checkmark$	(c) $-8x^3 - x^2 - 44x + 4\checkmark$	(d) $\frac{-52x+56}{-6}\mathbf{X}$	(e) $\frac{8x+32}{15x}\mathbf{X}$	(f) $\frac{16x^3+48x^2+30x}{-8x-2}\mathbf{X}$
	$ii : \mathbf{I}$	(a) $-36x + 30\mathbf{X}$	(b) $-24x^2 + 11x + 12\checkmark$	(c) $x^3 - 13x^2 + 55x - 76\mathbf{X}$	(d) $\frac{36x+81}{-42}\mathbf{X}$	(e) $\frac{20x-10}{-12x}\mathbf{X}$	(f) $\frac{33x^3-84x^2+53x}{-56x+21}\checkmark$
	$iii : \mathbf{N}$	(a) $-2x - 26\checkmark$	(b) $-4x^2 + 20x\mathbf{X}$	(c) $-8x^3 - 73x^2 - 76x - 68\checkmark$	(d) $\frac{-87x-87}{36}\mathbf{X}$	(e) $\frac{-24x-84}{-18x}\checkmark$	(f) $\frac{4x^3+4x^2-7x}{2x-8}\checkmark$
	$iv : \mathbf{E}$	(a) $-12x + 21\checkmark$	(b) $9x^2 - 27x + 108\mathbf{X}$	(c) $27x^3 + 38x^2 + 84x - 28\mathbf{X}$	(d) $\frac{8x-48}{16}\mathbf{X}$	(e) $\frac{16x+64}{62x}\checkmark$	(f) $\frac{16x^3-32x^2+17x}{21x+6}\mathbf{X}$
κ	$i : \mathbf{W}$	(a) $19x - 47\checkmark$	(b) $18x^2 - 27x + 39\checkmark$	(c) $64x^3 + 23x^2 + 42x - 8\checkmark$	(d) $\frac{110x+121}{21}\checkmark$	(e) $\frac{24x+12}{50x}\checkmark$	(f) $\frac{9x^3-12x^2+12x}{-2x-12}\mathbf{X}$
	$ii : \mathbf{I}$	(a) $50x\mathbf{X}$	(b) $-32x^2 - 114x - 66\checkmark$	(c) $-64x^3 + 191x^2 - 202x + 39\mathbf{X}$	(d) $\frac{-44x-62}{-8}\mathbf{X}$	(e) $\frac{36x+24}{-14x}\mathbf{X}$	(f) $\frac{4x^3-24x^2+39x}{-3x-6}\checkmark$
	$iii : \mathbf{N}$	(a) $-8x + 23\checkmark$	(b) $-14x^2 - 40x + 22\mathbf{X}$	(c) $8x^3 - 76x^2 + 38x - 5\checkmark$	(d) $\frac{20x-82}{10}\mathbf{X}$	(e) $\frac{6x-30}{-35x}\checkmark$	(f) $\frac{36x^3-84x^2+47x}{-8x+10}\checkmark$
	$iv : \mathbf{D}$	(a) $61x + 19\checkmark$	(b) $-12x^2 + 21x + 33\mathbf{X}$	(c) $-x^3 - 43x^2 + 2x + 7\mathbf{X}$	(d) $\frac{94x+50}{42}\mathbf{X}$	(e) $\frac{32x-96}{-72x}\checkmark$	(f) $\frac{x^3+2x^2}{x-9}\checkmark$
λ	$i : \mathbf{U}$	(a) $62x + 45\checkmark$	(b) $-36x^2 + 70x + 54\mathbf{X}$	(c) $64x^3 - 112x^2 + 92x - 26\checkmark$	(d) $\frac{43x+59}{-24}\checkmark$	(e) $\frac{4x-20}{40x}\mathbf{X}$	(f) $\frac{16x^3-48x^2+32x}{-x+3}\mathbf{X}$
	$ii : \mathbf{S}$	(a) $6x + 15\mathbf{X}$	(b) $12x^2 + 19x + 9\checkmark$	(c) $x^3 + 11x^2 + 46x + 63\checkmark$	(d) $\frac{20x+28}{-30}\mathbf{X}$	(e) $\frac{-6x+18}{90x}\mathbf{X}$	(f) $\frac{49x^3+70x^2+27x}{x-7}\checkmark$
	$iii : \mathbf{E}$	(a) $-5x + 12\checkmark$	(b) $-10x^2 - 73x - 59\mathbf{X}$	(c) $-64x^3 + 47x^2 + 4x - 63\mathbf{X}$	(d) $\frac{-74x-14}{-10}\mathbf{X}$	(e) $\frac{3x+24}{-4x}\checkmark$	(f) $\frac{25x^3-50x^2+29x}{48x+6}\mathbf{X}$
	$iv : \mathbf{R}$	(a) $-36x + 20\checkmark$	(b) $-12x^2 - 24x + 20\checkmark$	(c) $-8x^3 - 13x^2 - 18x - 37\checkmark$	(d) $\frac{33x+9}{10}\mathbf{X}$	(e) $\frac{-6x-96}{-60x}\checkmark$	(f) $\frac{9x^3+24x^2+21x}{-35x-20}\mathbf{X}$
μ	$i : \mathbf{B}$	(a) $-8x - 44\checkmark$	(b) $12x^2 + 51x - 43\checkmark$	(c) $8x^3 + 47x^2 + 92x + 60\mathbf{X}$	(d) $\frac{64x-64}{-32}\mathbf{X}$	(e) $\frac{-3x-48}{32x}\mathbf{X}$	(f) $\frac{x^3-4x^2-2x}{6x+6}\mathbf{X}$
	$ii : \mathbf{O}$	(a) $-22x + 21\checkmark$	(b) $28x^2 + 58x - 50\mathbf{X}$	(c) $x^3 - 45x^2 + 15x - 28\checkmark$	(d) $\frac{14x+20}{6}\mathbf{X}$	(e) $\frac{15x+24}{26x}\checkmark$	(f) $\frac{9x^3-6x^2+2x}{-3x-6}\mathbf{X}$
	$iii : \mathbf{S}$	(a) $27x + 12\mathbf{X}$	(b) $25x^2 - 61x - 42\checkmark$	(c) $-27x^3 - 28x^2 - 5x - 5\checkmark$	(d) $\frac{8x-33}{15}\mathbf{X}$	(e) $\frac{-4x-8}{12x}\mathbf{X}$	(f) $\frac{16x^3-8x^2+4x}{-6x-48}\checkmark$
	$iv : \mathbf{S}$	(a) $-20x + 5\mathbf{X}$	(b) $-48x^2 + 80x - 38\checkmark$	(c) $-64x^3 + 32x^2 - 128x - 17\checkmark$	(d) $\frac{-64x+50}{24}\mathbf{X}$	(e) $\frac{12x+6}{-13x}\mathbf{X}$	(f) $\frac{x^3-2x^2+5x}{3x+5}\checkmark$

Simplifying an Expression (solutions)

ν	$i : \mathbf{G}$	(a) $-69x - 13\checkmark$	(b) $12x^2 + 41x + 49\checkmark$	(c) $-x^3 - 12x^2 + 15x - 10\mathbf{X}$	(d) $\frac{-48x-62}{16}\mathbf{X}$	(e) $\frac{-28x-48}{-4x}\checkmark$	(f) $\frac{x^3-2x^2-8x}{-36x+20}\checkmark$
	$ii : \mathbf{O}$	(a) $-32x - 76\checkmark$	(b) $-60x^2 + 147x - 72\mathbf{X}$	(c) $27x^3 - 10x^2 + 52x + 7\checkmark$	(d) $\frac{7x+28}{49}\mathbf{X}$	(e) $\frac{60x-75}{-37x}\checkmark$	(f) $\frac{4x^3+20x^2+28x}{8x-4}\mathbf{X}$
	$iii : \mathbf{A}$	(a) $-58x + 2\checkmark$	(b) $-6x^2 + 4x + 65\mathbf{X}$	(c) $64x^3 + 12x^2 + 36x - 3\mathbf{X}$	(d) $\frac{-6x-74}{14}\mathbf{X}$	(e) $\frac{-6x+36}{38x}\mathbf{X}$	(f) $\frac{x^3-10x^2+27x}{7x+49}\mathbf{X}$
	$iv : \mathbf{L}$	(a) $4x - 10\checkmark$	(b) $-8x^2 - 20x + 84\checkmark$	(c) $-x^3 + 5x^2 - 28\checkmark$	(d) $\frac{64x-78}{-28}\mathbf{X}$	(e) $\frac{-45x+72}{-40x}\mathbf{X}$	(f) $\frac{49x^3-14x^2+7x}{-12x+12}\mathbf{X}$
	$i : \mathbf{S}$	(a) $2x + 6\mathbf{X}$	(b) $14x^2 + 9x - 61\checkmark$	(c) $-x^3 - 30x^2 - 24x + 7\checkmark$	(d) $\frac{99x+67}{12}\mathbf{X}$	(e) $\frac{-12x+24}{-26x}\mathbf{X}$	(f) $\frac{9x^3-12x^2+x}{-8x+4}\checkmark$
ξ	$ii : \mathbf{A}$	(a) $-36x + 32\checkmark$	(b) $-100x^2 - 62x - 14\mathbf{X}$	(c) $8x^3 - 61x^2 + 62x - 17\mathbf{X}$	(d) $\frac{30x-6}{12}\mathbf{X}$	(e) $\frac{-12x-36}{-38x}\mathbf{X}$	(f) $\frac{36x^3+12x^2+6x}{24x-4}\mathbf{X}$
	$iii : \mathbf{V}$	(a) $12x - 6\checkmark$	(b) $-40x^2 + 52x - 9\checkmark$	(c) $-8x^3 - 61x^2 - 62x - 17\checkmark$	(d) $\frac{80x^3+40}{32}\checkmark$	(e) $\frac{-18x-24}{26x}\mathbf{X}$	(f) $\frac{49x^3+126x^2-85x}{-2x-5}\mathbf{X}$
	$iv : \mathbf{E}$	(a) $9x - 18\checkmark$	(b) $24x^2 + 136x + 112\mathbf{X}$	(c) $27x^3 - 37x^2 + 105x - 35\mathbf{X}$	(d) $\frac{3x+36}{-9}\mathbf{X}$	(e) $\frac{-4x-32}{-10x}\checkmark$	(f) $\frac{4x^3+4x^2+9x}{-6x+9}\mathbf{X}$
	$i : \mathbf{S}$	(a) $22x - 69\mathbf{X}$	(b) $-18x^2 - 4x + 36\checkmark$	(c) $8x^3 - 85x^2 + 96x - 36\checkmark$	(d) $\frac{95x+33}{12}\mathbf{X}$	(e) $\frac{-25x+5}{-6x}\mathbf{X}$	(f) $\frac{4x^3-4x^2}{-4x-14}\checkmark$
$\circ$	$ii : \mathbf{E}$	(a) $6x - 2\checkmark$	(b) $4x^2 + 24x + 27\mathbf{X}$	(c) $x^3 - 15x^2 - 6x - 17\mathbf{X}$	(d) $\frac{-48x+24}{10}\mathbf{X}$	(e) $\frac{-8x+24}{7x}\checkmark$	(f) $\frac{16x^3-8x^2+7x}{56x-63}\mathbf{X}$
	$iii : \mathbf{A}$	(a) $26x - 11\checkmark$	(b) $-60x^2 - 11x + 3\mathbf{X}$	(c) $27x^3 - 70x^2 + 52x - 12\mathbf{X}$	(d) $\frac{-11x-13}{12}\mathbf{X}$	(e) $\frac{3x+3}{-5x}\mathbf{X}$	(f) $\frac{64x^3-16x^2-2x}{48x-16}\mathbf{X}$
	$iv : \mathbf{T}$	(a) $6x + 7\mathbf{X}$	(b) $-96x^2 + 18x + 9\checkmark$	(c) $-27x^3 - 31x^2 - x - 5\checkmark$	(d) $\frac{3x+33}{9}\mathbf{X}$	(e) $\frac{-28x-63}{9x}\checkmark$	(f) $\frac{36x^3+24x^2+8x}{20x-5}\mathbf{X}$
	$i : \mathbf{D}$	(a) $18x + 33\checkmark$	(b) $30x^2 + 123x\mathbf{X}$	(c) $8x^3 - 61x^2 + 24x - 36\mathbf{X}$	(d) $\frac{72x-16}{20}\mathbf{X}$	(e) $\frac{-3x-3}{-22x}\checkmark$	(f) $\frac{4x^3-16x^2+12x}{24x+12}\checkmark$
π	$ii : \mathbf{A}$	(a) $-24x - 10\checkmark$	(b) $-8x^2 + 24x - 22\mathbf{X}$	(c) $-x^3 - 33x^2 - 15x\mathbf{X}$	(d) $\frac{-30x-6}{48}\mathbf{X}$	(e) $\frac{-3x-36}{60x}\mathbf{X}$	(f) $\frac{9x^3+6x^2+5x}{-3x+21}\mathbf{X}$
	$iii : \mathbf{T}$	(a) $-18x - 10\mathbf{X}$	(b) $-45x^2 - 96x - 12\checkmark$	(c) $-8x^3 - 37x^2 - 26x - 5\checkmark$	(d) $\frac{5}{-15}\mathbf{X}$	(e) $\frac{18x-24}{-20x}\checkmark$	(f) $\frac{25x^3+20x^2+3x}{-63x+14}\checkmark$
	$iv : \mathbf{E}$	(a) $-52x + 87\checkmark$	(b) $18x^2 - 45x + 45\mathbf{X}$	(c) $-64x^3 - 100x^2 - 52x - 9\mathbf{X}$	(d) $\frac{23x-58}{20}\mathbf{X}$	(e) $\frac{-70x+84}{-15x}\checkmark$	(f) $\frac{49x^3-84x^2+38x}{15x+6}\mathbf{X}$
	$i : \mathbf{E}$	(a) $-55x + 15\checkmark$	(b) $27x^2 - 96x + 45\mathbf{X}$	(c) $-27x^3 + 45x^2 - 72x - 28\mathbf{X}$	(d) $\frac{9x+27}{30}\mathbf{X}$	(e) $\frac{-72x+24}{-102x}\checkmark$	(f) $\frac{36x^3-12x^2}{5x-1}\mathbf{X}$
ρ	$ii : \mathbf{A}$	(a) $-31x + 5\checkmark$	(b) $-15x^2 - 58x - 64\mathbf{X}$	(c) $x^3 - x^2 - 9x - 8\mathbf{X}$	(d) $\frac{-69x+27}{18}\mathbf{X}$	(e) $\frac{81x-45}{9x}\mathbf{X}$	(f) $\frac{16x^3+24x^2+13x}{9x-13}\mathbf{X}$
	$iii : \mathbf{S}$	(a) $9x + 18\mathbf{X}$	(b) $-27x^2 + 63x + 60\checkmark$	(c) $8x^3 + 20x^2 + 94x + 2\checkmark$	(d) $\frac{-48x-90}{-36}\mathbf{X}$	(e) $\frac{-18x-90}{-39x}\mathbf{X}$	(f) $\frac{81x^3+18x^2-7x}{x+1}\checkmark$
	$iv : \mathbf{T}$	(a) $33x - 15\mathbf{X}$	(b) $24x^2 + 24x - 40\checkmark$	(c) $-x^3 - 75x^2 + 6x + 7\checkmark$	(d) $\frac{x-32}{-35}\mathbf{X}$	(e) $\frac{3x-5}{-5x}\checkmark$	(f) $\frac{4x^3-16x^2+10x}{45x+15}\checkmark$
	$i : \mathbf{R}$	(a) $-40x - 80\checkmark$	(b) $-100x^2 - 72x - 8\checkmark$	(c) $8x^3 + 47x^2 + 104x + 48\checkmark$	(d) $\frac{-26x+91}{20}\mathbf{X}$	(e) $\frac{-36x-36}{21x}\checkmark$	(f) $\frac{36x^3+96x^2+70x}{-63x+45}\mathbf{X}$
σ	$ii : \mathbf{O}$	(a) $53x + 50\checkmark$	(b) $-27x^2 - 26x + 6\mathbf{X}$	(c) $x^3 + 8x^2 + 56x + 60\checkmark$	(d) $\frac{-50x+122}{-24}\mathbf{X}$	(e) $\frac{2x-8}{-2x}\checkmark$	(f) $\frac{9x^3-12x^2}{-4x+16}\mathbf{X}$
	$iii : \mathbf{O}$	(a) $16x + 9\checkmark$	(b) $-12x^2 - 60x + 27\mathbf{X}$	(c) $x^3 - 10x^2 + 29x - 28\checkmark$	(d) $\frac{-36x+78}{-36}\mathbf{X}$	(e) $\frac{-28x-21}{87x}\checkmark$	(f) $\frac{4x^3+8x^2+3x}{x^3-7x+1}\mathbf{X}$
	$iv : \mathbf{M}$	(a) $30x - 42\checkmark$	(b) $64x^2 - 28x + 18\mathbf{X}$	(c) $-64x^3 - 193x^2 - 188x - 68\checkmark$	(d) $\frac{-15x+78}{-14}\mathbf{X}$	(e) $\frac{-16x-60}{20x}\mathbf{X}$	(f) $\frac{x^3-6x^2+11x}{-6x+3}\checkmark$
	$i : \mathbf{T}$	(a) $74x + 44\mathbf{X}$	(b) $14x^2 - 39x - 11\checkmark$	(c) $64x^3 - 97x^2 - 72x - 37\checkmark$	(d) $\frac{5x+35}{-25}\mathbf{X}$	(e) $\frac{6x+36}{-33x}\checkmark$	(f) $\frac{x^3-4x^2-x}{x+8}\checkmark$
τ	$ii : \mathbf{R}$	(a) $2x + 19\checkmark$	(b) $-12x^2 + 62x + 2\checkmark$	(c) $x^3 - 7x^2 - 9x - 10\checkmark$	(d) $\frac{-26x+130}{-10}\mathbf{X}$	(e) $\frac{35x+90}{-39x}\checkmark$	(f) $\frac{x^3-4x^2+3x}{-28x-14}\mathbf{X}$
	$iii : \mathbf{E}$	(a) $-52x + 41\checkmark$	(b) $-12x^2 + 15x + 29\mathbf{X}$	(c) $-27x^3 + 11x^2 + 23x - 15\mathbf{X}$	(d) $\frac{4x-8}{8}\mathbf{X}$	(e) $\frac{-40x+80}{36x}\checkmark$	(f) $\frac{64x^3+16x^2-4x}{-3x+3}\mathbf{X}$
	$iv : \mathbf{E}$	(a) $40x + 4\checkmark$	(b) $-32x^2 + 93x - 27\mathbf{X}$	(c) $27x^3 - 22x^2 + 23x\mathbf{X}$	(d) $\frac{-2x+28}{30}\mathbf{X}$	(e) $\frac{20x+8}{6x}\checkmark$	(f) $\frac{4x^3-20x^2+28x}{4x+32}\mathbf{X}$
	$i : \mathbf{B}$	(a) $-9x - 15\checkmark$	(b) $72x^2 - 52x + 2\checkmark$	(c) $-8x^3 - 52x^2 - 76x - 89\mathbf{X}$	(d) $\frac{-102x-15}{-18}\mathbf{X}$	(e) $\frac{-24x+72}{57x}\mathbf{X}$	(f) $\frac{16x^3+40x^2+18x}{6x-8}\mathbf{X}$
v	$ii : \mathbf{O}$	(a) $-4x - 69\checkmark$	(b) $8x^2 + 20x + 17\mathbf{X}$	(c) $-8x^3 + 39x^2 - 66x + 39\checkmark$	(d) $\frac{30x-15}{9}\mathbf{X}$	(e) $\frac{-56x+24}{33x}\checkmark$	(f) $\frac{4x^3+8x^2+6x}{20x-5}\mathbf{X}$
	$iii : \mathbf{D}$	(a) $-6x - 7\checkmark$	(b) $48x^2 - 84x + 32\mathbf{X}$	(c) $27x^3 - 97x^2 + 57x - 36\mathbf{X}$	(d) $\frac{17x+5}{20}\mathbf{X}$	(e) $\frac{-4x-16}{-4x}\checkmark$	(f) $\frac{x^3-18x^2+77x}{28x+7}\mathbf{X}$
	$iv : \mathbf{Y}$	(a) $2x - 4\checkmark$	(b) $-4x^2 + 57x - 18\mathbf{X}$	(c) $64x^3 - 73x^2 + 42x - 10\checkmark$	(d) $\frac{-47x-91}{-15}\checkmark$	(e) $\frac{-12x+75}{45x}\checkmark$	(f) $\frac{x^3-8x^2+11x}{-8x+4}\checkmark$
	$i : \mathbf{F}$	(a) $-6x - 34\checkmark$	(b) $12x^2 - 29x - 34\checkmark$	(c) $-8x^3 - 25x^2 - 108x - 28\mathbf{X}$	(d) $\frac{-71x+54}{-21}\mathbf{X}$	(e) $\frac{-20x-64}{-28x}\mathbf{X}$	(f) $\frac{36x^3-12x^2-6x}{-48x-12}\checkmark$
ϕ	$ii : \mathbf{E}$	(a) $-39x + 37\checkmark$	(b) $72x^2 - 72x - 43\mathbf{X}$	(c) $64x^3 + 191x^2 + 190x + 63\mathbf{X}$	(d) $\frac{69x-38}{-72}\mathbf{X}$	(e) $\frac{80x-20}{-21x}\checkmark$	(f) $\frac{4x^3+16x^2+18x}{4x+8}\mathbf{X}$
	$iii : \mathbf{E}$	(a) $-26x - 43\checkmark$	(b) $-12x^2 + 85x + 69\mathbf{X}$	(c) $-27x^3 - 97x^2 - 121x - 52\mathbf{X}$	(d) $\frac{48x+48}{-18}\mathbf{X}$	(e) $\frac{12x+60}{-6x}\checkmark$	(f) $\frac{9x^3+12x^2-2x}{28x+7}\mathbf{X}$
	$iv : \mathbf{T}$	(a) $13x - 36\mathbf{X}$	(b) $-32x^2 + 85x - 52\checkmark$	(c) $64x^3 - 196x^2 + 224x - 128\checkmark$	(d) $\frac{-27x+70}{-42}\mathbf{X}$	(e) $\frac{-4x+8}{-8x}\checkmark$	(f) $\frac{4x^3-12x^2+7x}{-10x-10}\checkmark$
	$i : \mathbf{W}$	(a) $9x - 12\checkmark$	(b) $60x^2 + 134x + 98\checkmark$	(c) $27x^3 - 112x^2 + 116x - 113\checkmark$	(d) $\frac{-48x+16}{12}\checkmark$	(e) $\frac{-24x-36}{4x}\checkmark$	(f) $\frac{16x^3+40x^2+31x}{-2x-6}\mathbf{X}$
χ	$ii : \mathbf{A}$	(a) $24x - 36\checkmark$	(b) $24x^2 + 33x - 37\mathbf{X}$	(c) $27x^3 - 109x^2 + 136x - 80\mathbf{X}$	(d) $\frac{-102x-42}{45}\mathbf{X}$	(e) $\frac{-36x+4}{-10x}\mathbf{X}$	(f) $\frac{x^3-4x^2-3x}{-6x+24}\mathbf{X}$
	$iii : \mathbf{L}$	(a) $2x + 9\checkmark$	(b) $96x^2 - 27x - 48\checkmark$	(c) $-27x^3 - 103x^2 - 50x - 9\checkmark$	(d) $\frac{-41x-61}{-20}\mathbf{X}$	(e) $\frac{-6x-18}{x}\mathbf{X}$	(f) $\frac{x^3-6x^2+12x}{-2x-10}\mathbf{X}$
	$iv : \mathbf{K}$	(a) $30x - 10\checkmark$	(b) $36x^2 - 105x - 33\mathbf{X}$	(c) $-27x^3 + 18x^2 - 33x - 15\checkmark$	(d) $\frac{-87x-41}{12}\mathbf{X}$	(e) $\frac{-12x+12}{22x}\mathbf{X}$	(f) $\frac{16x^3+24x^2+11x}{-40x-30}\mathbf{X}$
	$i : \mathbf{B}$	(a) $-38x - 4\checkmark$	(b) $-32x^2 - 88x + 20\checkmark$	(c) $8x^3 - 13x^2 + 76x - 48\mathbf{X}$	(d) $\frac{22x+82}{12}\mathbf{X}$	(e) $\frac{4x-21}{-7x}\mathbf{X}$	(f) $\frac{25x^3-20x^2+2x}{3x+3}\mathbf{X}$
ψ	$ii : \mathbf{A}$	(a) $10x + 23\checkmark$	(b) $20x^2 - 69x - 49\mathbf{X}$	(c) $-64x^3 + 143x^2 - 126x - 54\mathbf{X}$	(d) $\frac{88x+16}{16}\mathbf{X}$	(e) $\frac{60x+60}{20x}\mathbf{X}$	(f) $\frac{16x^3+32x^2+11x}{6x-18}\mathbf{X}$
	$iii : \mathbf{B}$	(a) $-13x - 36\checkmark$	(b) $36x^2 - 45x + 6\checkmark$	(c) $-64x^3 - 193x^2 - 150x - 36\mathbf{X}$	(d) $\frac{154x-30}{-24}\mathbf{X}$	(e) $\frac{-16x+64}{22x}\mathbf{X}$	(f) $\frac{9x^3-18x^2+16x}{24x-3}\mathbf{X}$
	$iv : \mathbf{Y}$	(a) $10x - 10\checkmark$	(b) $48x^2 + 64x - 62\mathbf{X}$	(c) $64x^3 - 100x^2 + 72x - 44\checkmark$	(d) $\frac{76x-64}{32}\checkmark$	(e) $\frac{42x-48}{-4x}\checkmark$	(f) $\frac{36x^3-36x^2+4x}{3x-3}\checkmark$
	$i : \mathbf{U}$	(a) $-4x - 32\checkmark$	(b) $-27x^2 - 33x + 15\mathbf{X}$	(c) $-64x^3 - 16x^2 - 92x - 24\checkmark$	(d) $\frac{40x-40}{12}\mathbf{X}$	(e) $\frac{4x+4}{10x}\mathbf{X}$	(f) $\frac{16x^3+72x^2+80x}{34x+20}\mathbf{X}$
ω	$ii : \mathbf{N}$	(a) $-38x + 2\checkmark$	(b) $8x^2 - 22x - 30\mathbf{X}$	(c) $-x^3 - 4x^2 - 17x - 50\checkmark$	(d) $\frac{-39x-83}{-12}\mathbf{X}$	(e) $\frac{-x+21}{70x}\checkmark$	(f) $\frac{x^3+12x^2+40x}{-21x+21}\checkmark$
	$iii : \mathbf{I}$	(a) $19x + 14\mathbf{X}$	(b) $-24x^2 - 4x - 8\checkmark$	(c) $x^3 - 33x^2 + 39x - 8\mathbf{X}$	(d) $\frac{-36}{-24}\mathbf{X}$	(e) $\frac{14x+8}{-10x}\mathbf{X}$	(f) $\frac{4x^3+8x^2+x}{24x+18}\checkmark$
	$iv : \mathbf{T}$	(a) $-20x - 29\mathbf{X}$	(b) $-6x^2 + 30x + 57\checkmark$	(c) $x^3 - 45x^2 - 9x - 36\checkmark$	(d) $\frac{-80x-32}{28}\mathbf{X}$	(e) $\frac{42x-63}{-18x}\checkmark$	(f) $\frac{9x^3+42x^2+50x}{5x-5}\checkmark$